

Joint Attention-Field × PRF model

Distractor-driven shifts of population receptive
fields in human visual cortex

meeting with Dock Duncan & Jan Theeuwes

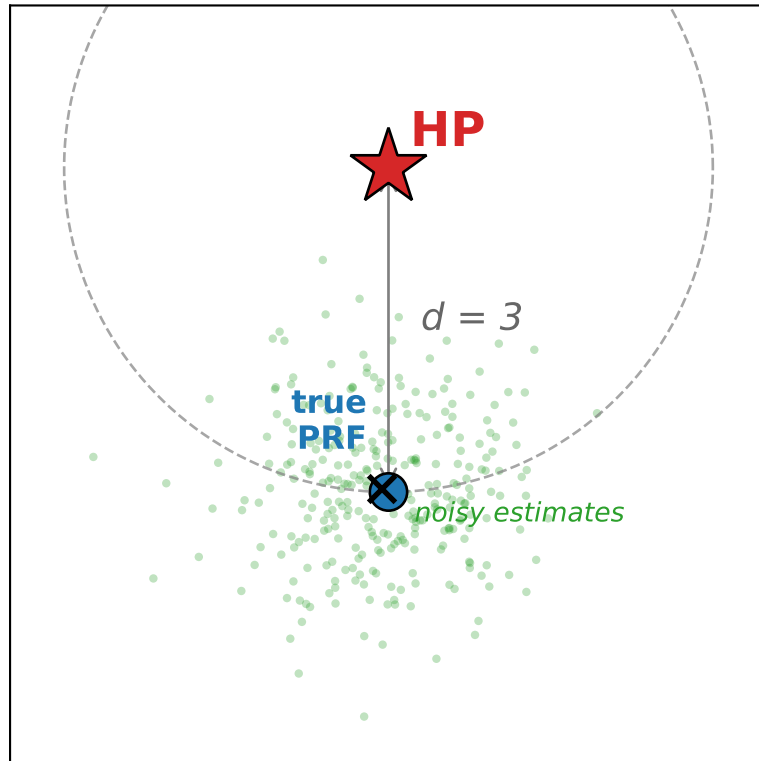
retsupp · 7T fMRI · 30 subjects · 4 HP-distractor blocks · sweeping bar PRF mapping

(a) Distance binning is biased

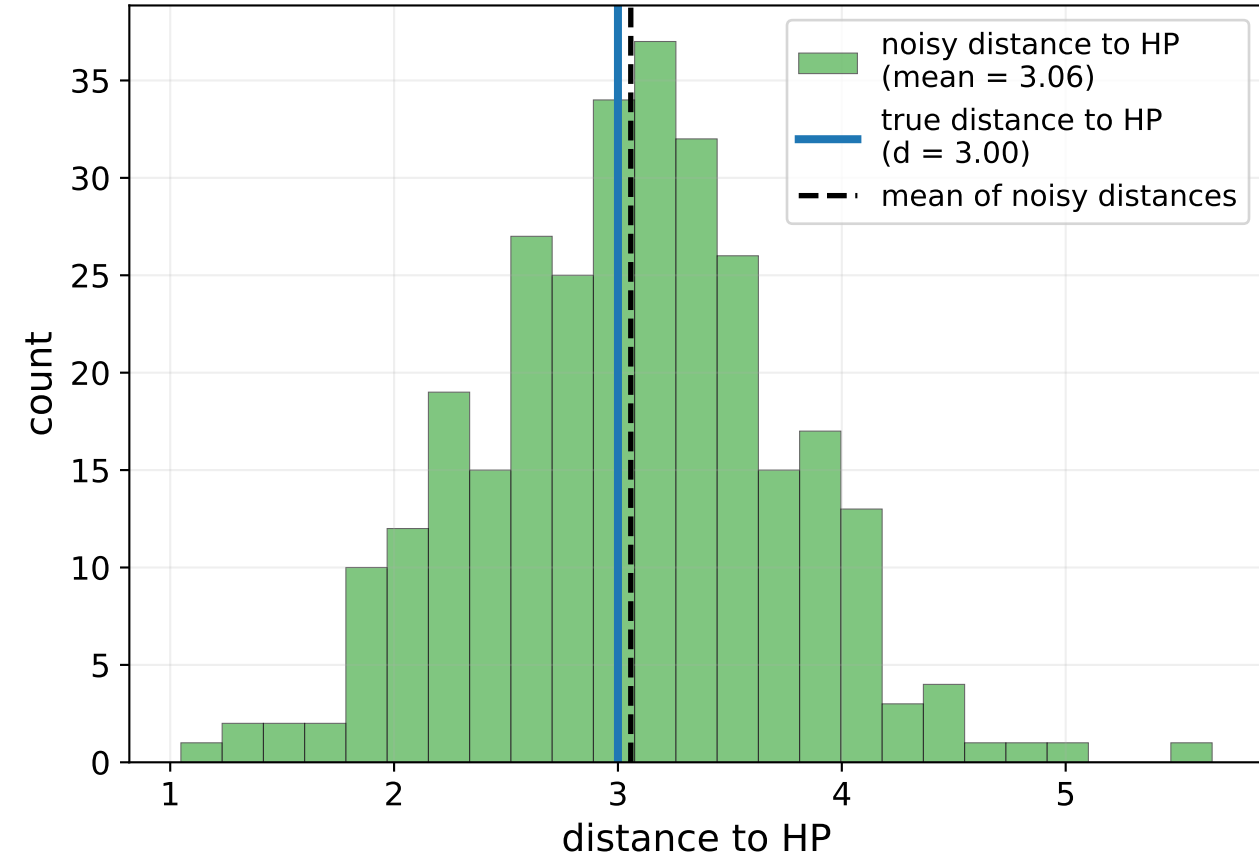
Random noise on PRF position pushes distance-to-HP up.

**(a) Random noise pushes "distance to HP" upward.
Hence "the PRF shifted AWAY from HP" can be a noise artifact.**

Take any voxel a distance d from HP.
Add isotropic noise to its PRF estimate.



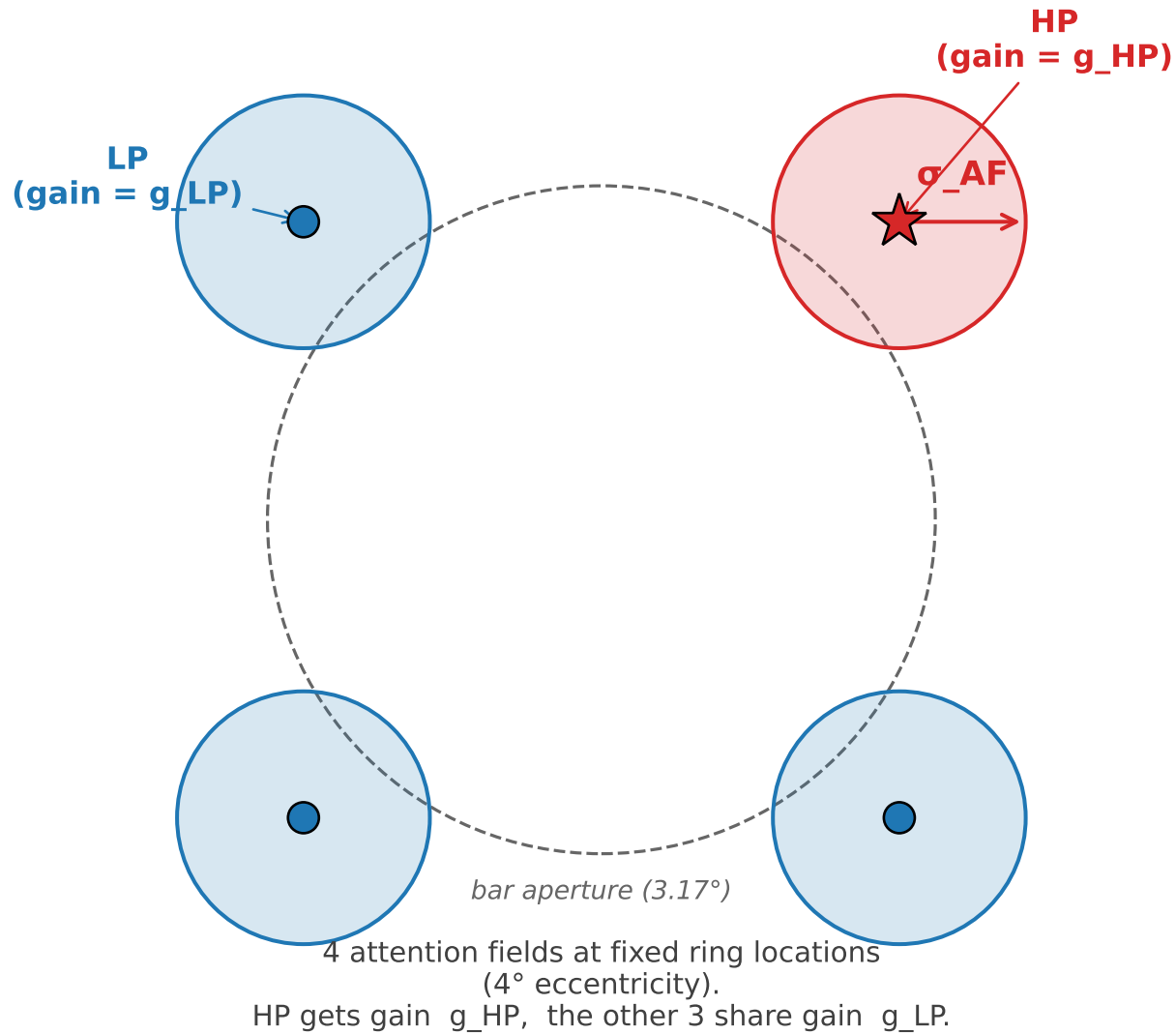
On AVERAGE, noisy distances $>$ true distance.
(Distance is convex — Jensen.)



(b) The model

4 attention fields $\times$ Gaussian voxel kernel; 3 shared parameters per ROI per subject.

(b) The model



Forward model

For each voxel v under HP condition C :

$$R_{v_C}(g) = M_C(g) \cdot S_v(g)$$

with the modulation field

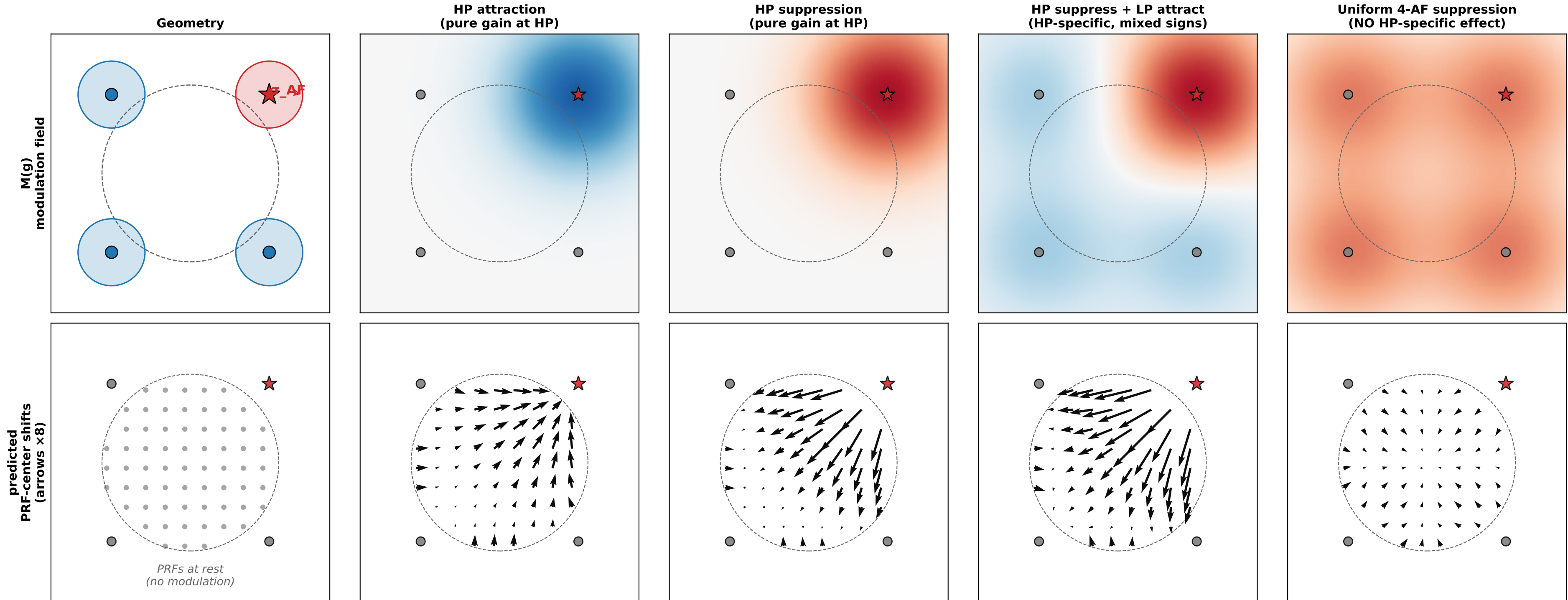
$$M_C(g) = 1 + g_{HP} \cdot A_{HP_C}(g) + g_{LP} \cdot \sum A_{LP}(g)$$

3 SHARED parameters per ROI:

σ_{AF}	AF size
g_{HP}	gain at HP location
g_{LP}	gain at LP locations

Negative gain $\Rightarrow$ suppression.

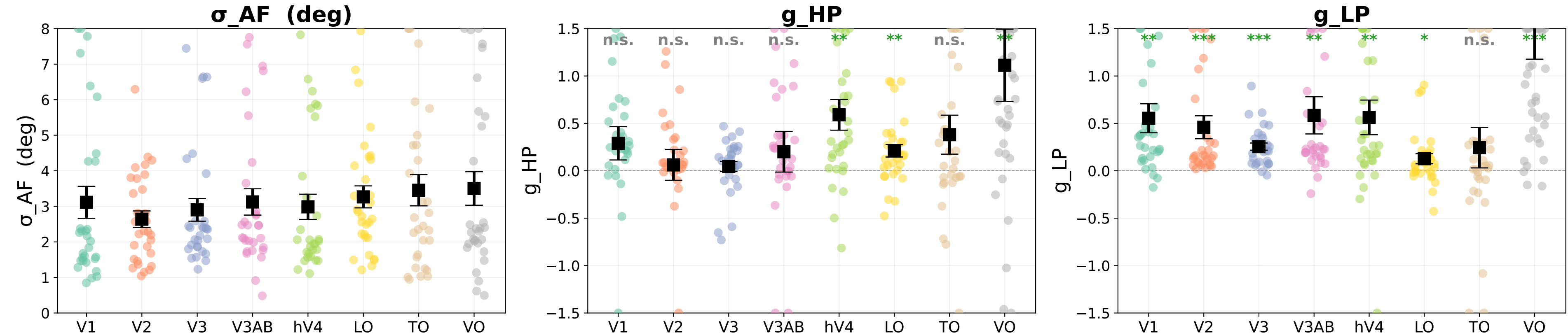
(b) What the model predicts at different parameter settings.
Top: modulation field $M(g)$ (RED = suppression, BLUE = attraction). **Bottom: predicted PRF-center shifts under HP=★.**



(c) Parameter estimates

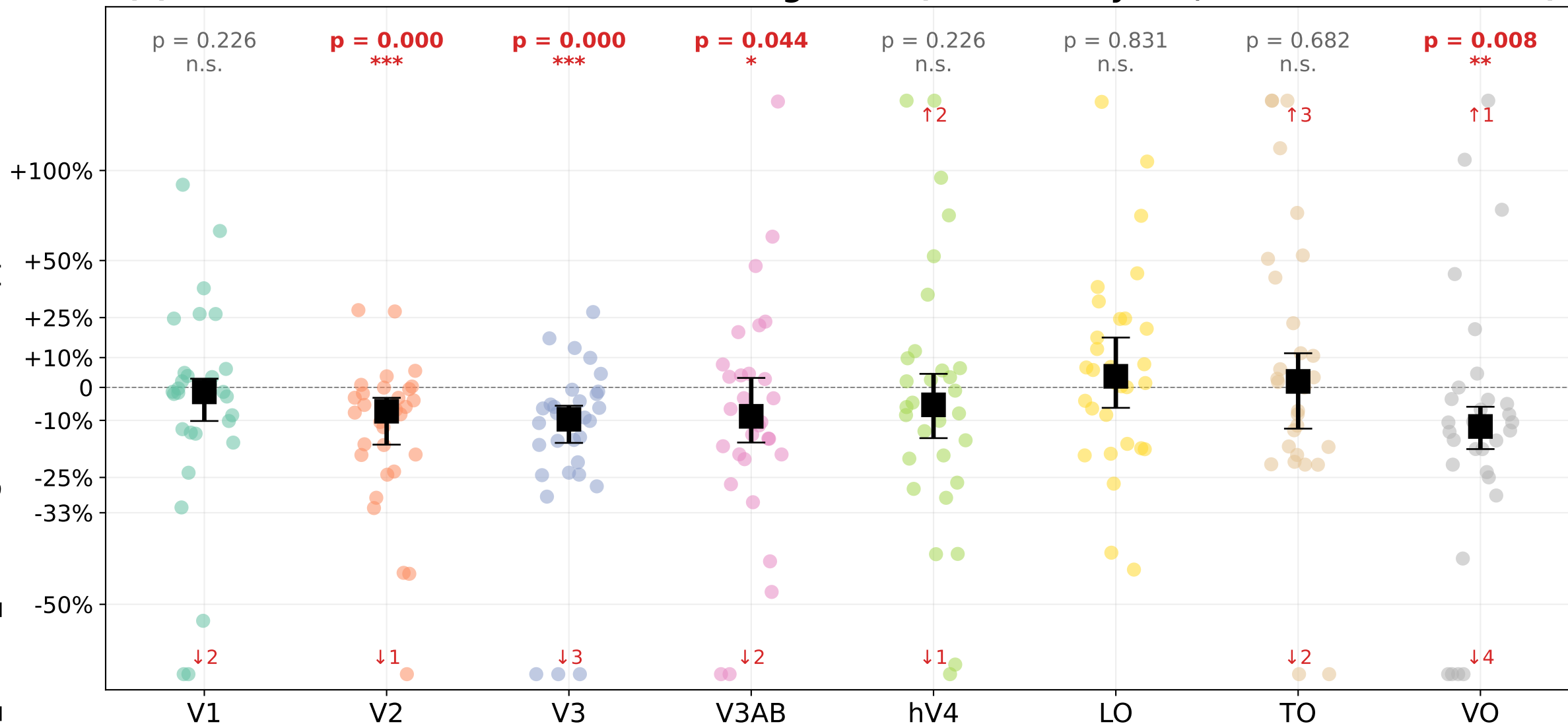
Per-ROI distributions and HP-vs-LP test ($N = 30$ subjects).

(c) AF parameter estimates per ROI (30 subjects, 240 fits) — dots = subjects, black square = mean $\pm$ SEM



M_{HP} / M_{LP} (negative $\Rightarrow$ HP suppressed more than LP)

(c) HP-vs-LP test — scale-invariant log-ratio (N = 30 subjects, Wilcoxon one-sided)



(d) What the model predicts per ROI

Per-voxel mean across 4 conditions vs HP=★ condition. Model on the left, data on the right.

(d) Group (median across all 30 subjects) — per-ROI predicted vs observed PRF-center shifts.

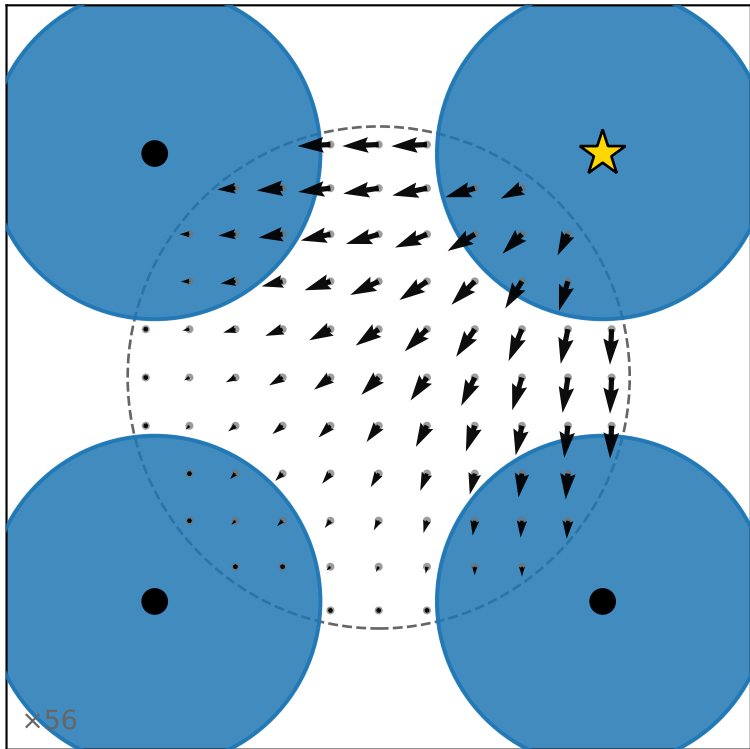
For each ROI: LEFT = model prediction, RIGHT = data.

Arrows: condition-specific shift (HP=★ – mean across 4 conditions for PRED;
mean-across-subjects of per-subject median shift for OBS).

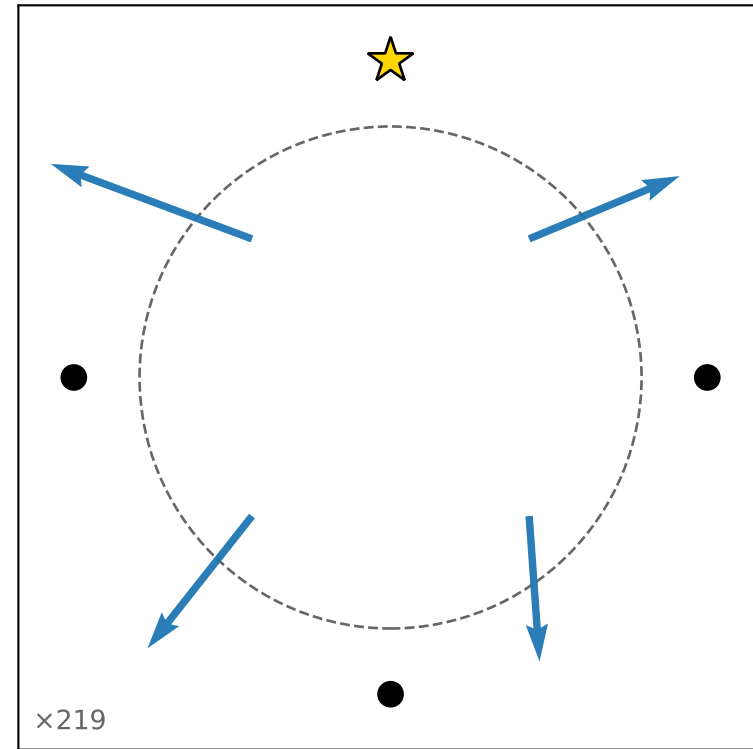
Circles: each AF (size = σ_{AF} , shading $\propto |\text{gain}|$, red = suppression, blue = attraction).

Arrow $\times N$ annotation in each panel = auto-scale (85th-ptile arrow $\approx 70\%$ of a bin width, so PRED and OBS are comparable in shape).

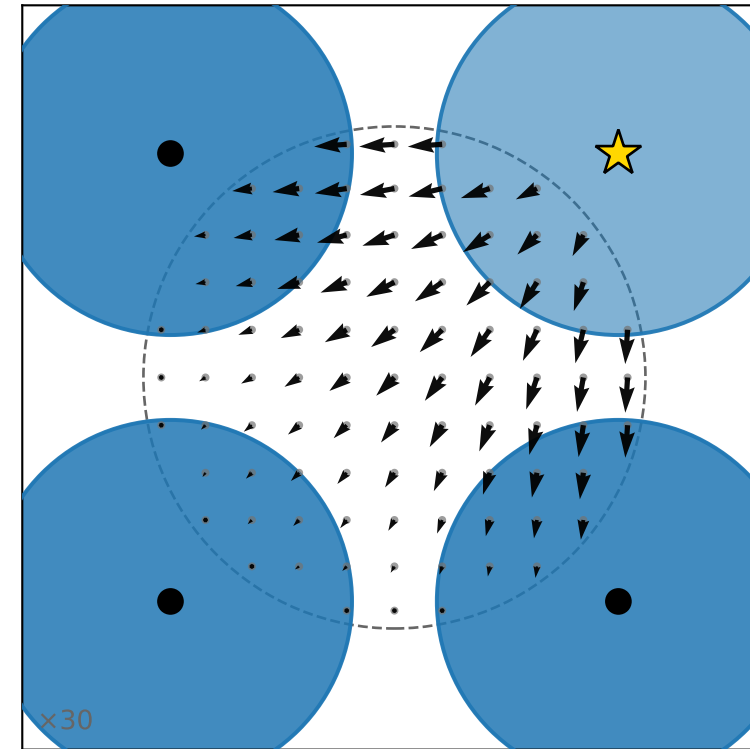
V1 PRED $\sigma=2.09$ $g_{HP}=+0.26$ $g_{LP}=+0.31$



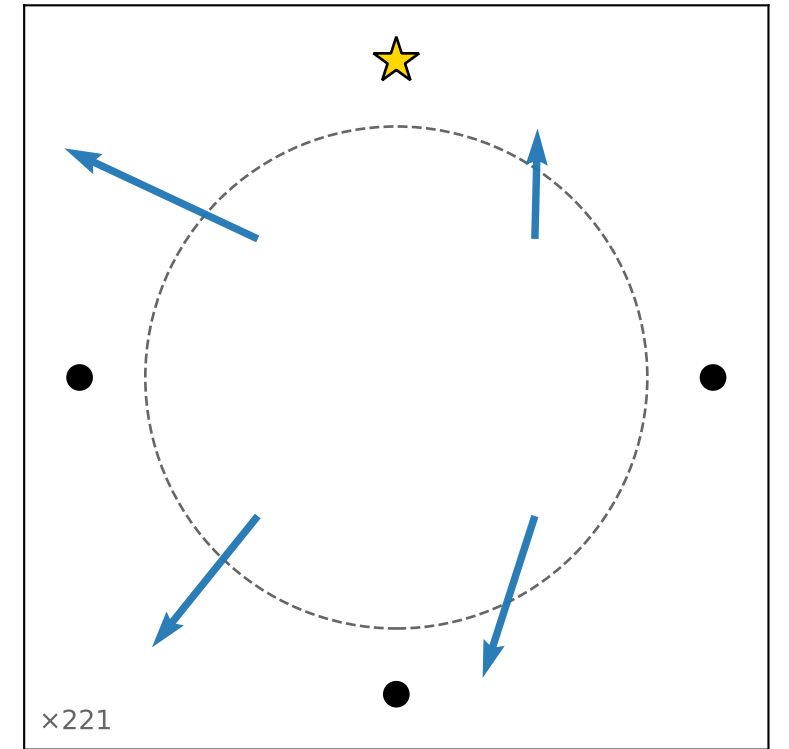
V1 DATA $n_{\text{subj}} = 28$



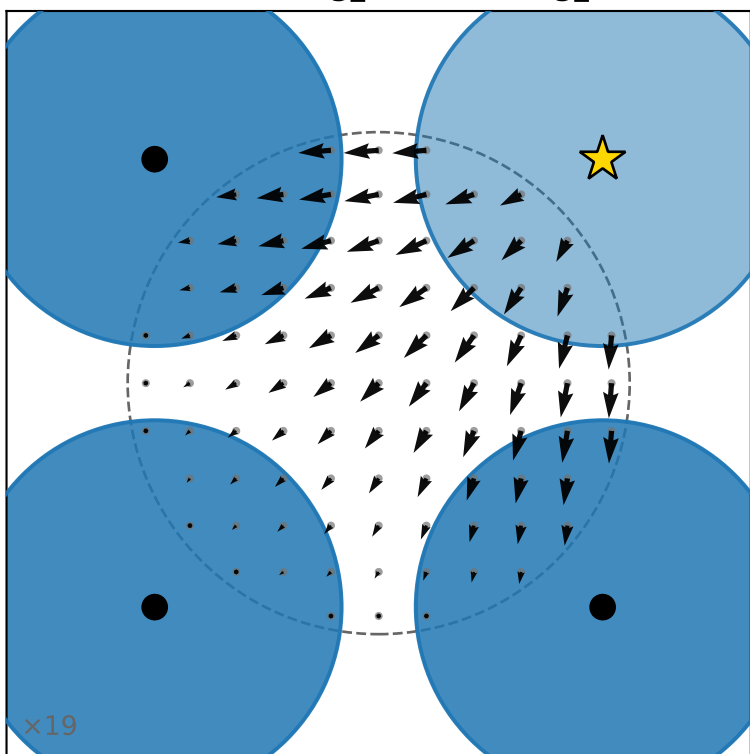
V2 PRED $\sigma=2.29$ $g_{HP}=+0.08$ $g_{LP}=+0.16$



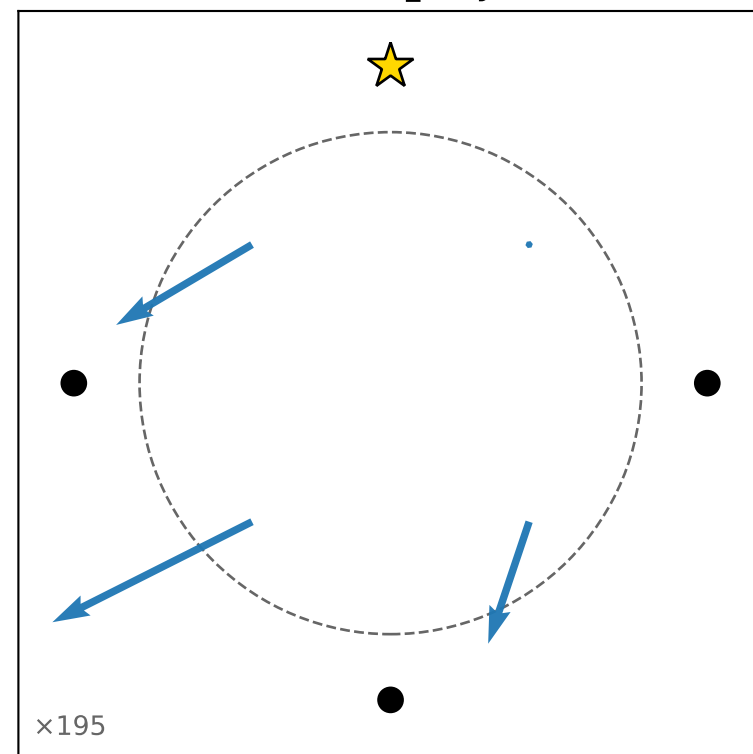
V2 DATA $n_{\text{subj}} = 28$



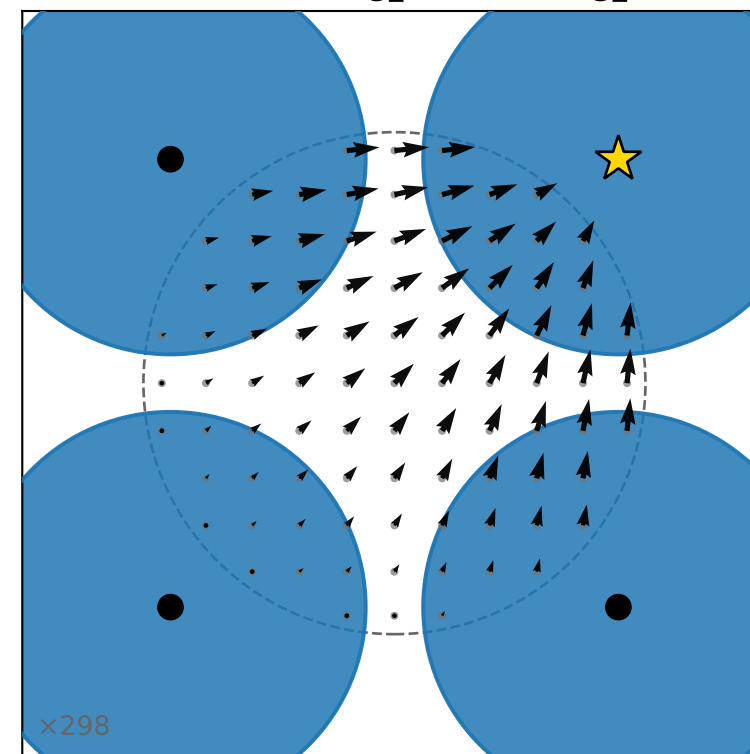
V3 PRED $\sigma=2.36$ $g_{HP}=+0.10$ $g_{LP}=+0.23$



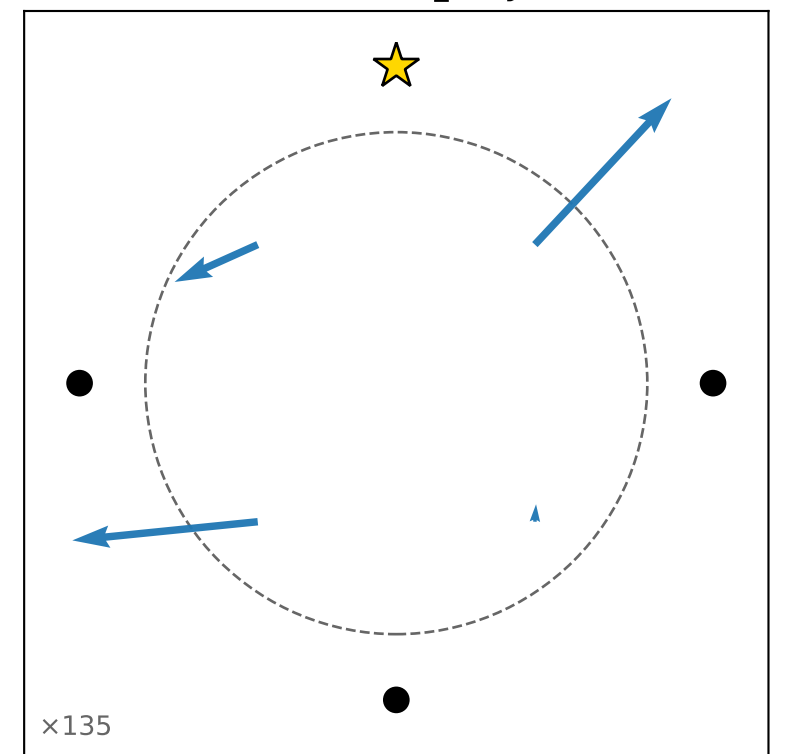
V3 DATA $n_{\text{subj}} = 28$



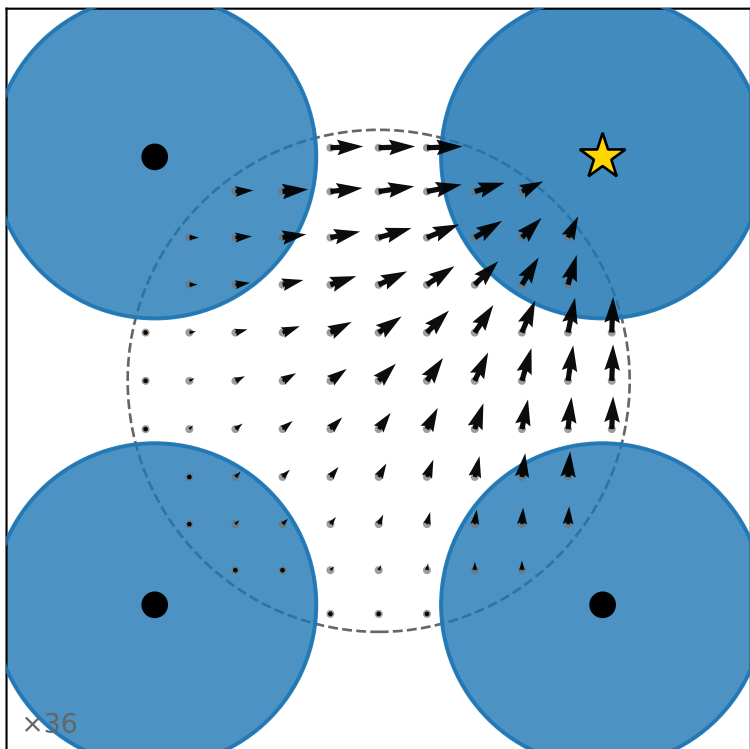
V3AB PRED $\sigma=2.47$ $g_{HP}=+0.24$ $g_{LP}=+0.23$



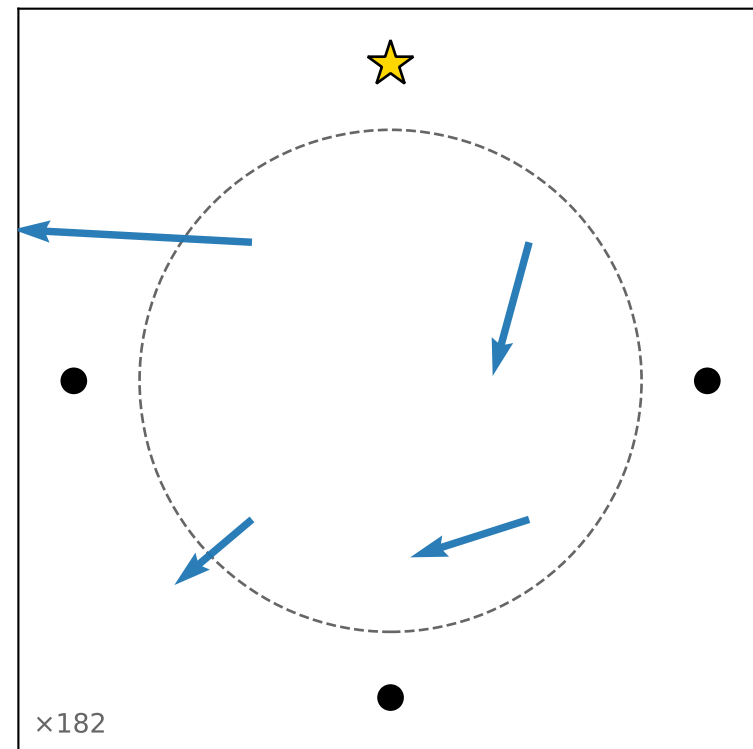
V3AB DATA $n_{\text{subj}} = 28$



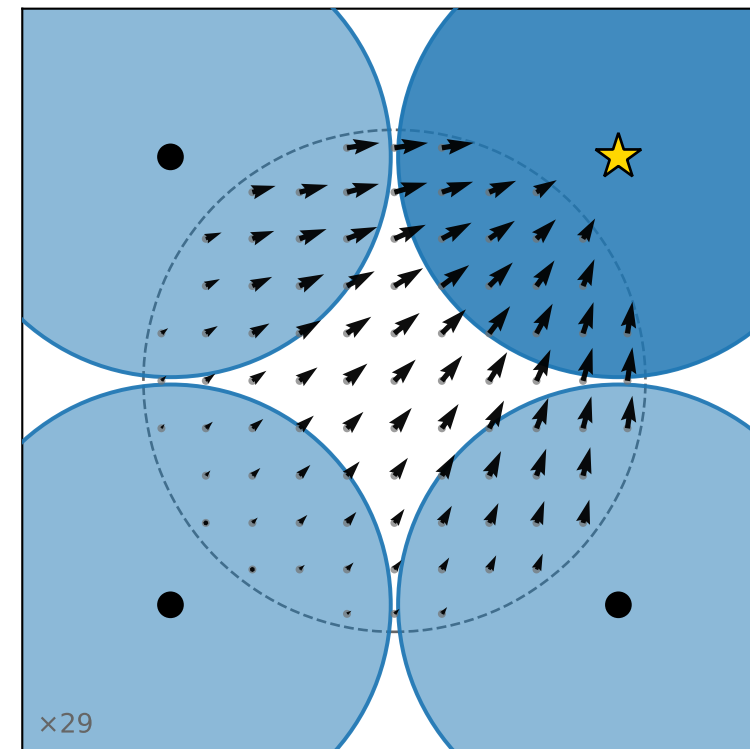
hV4 PRED $\sigma=2.04$ $g_{HP}=+0.32$ $g_{LP}=+0.24$



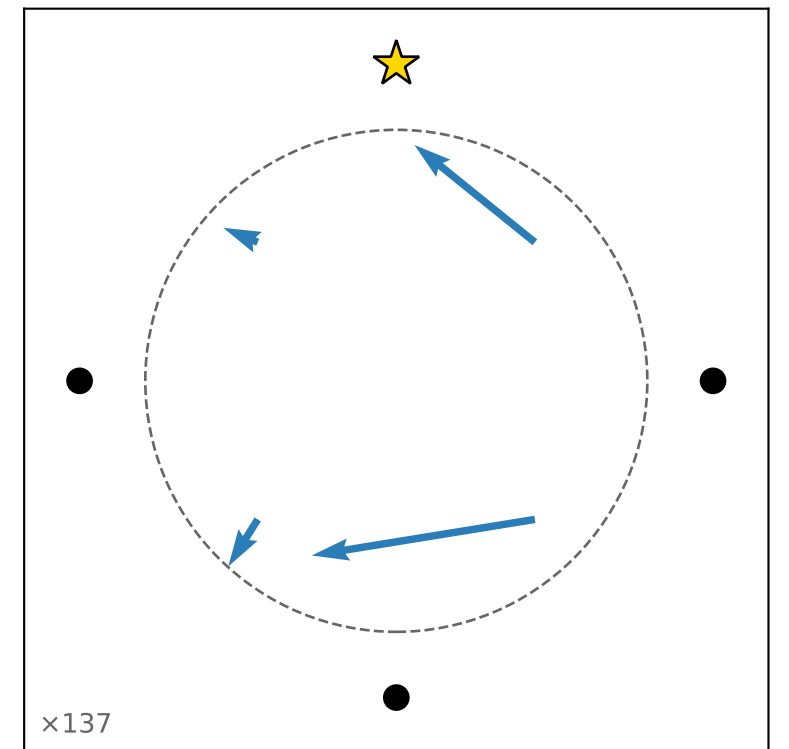
hV4 DATA $n_{\text{subj}} = 28$



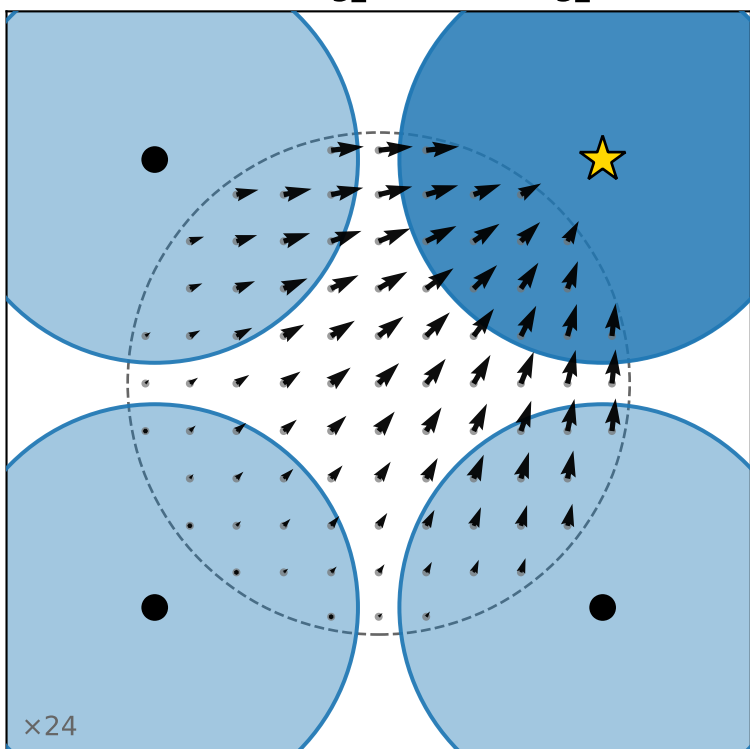
LO PRED $\sigma=2.78$ $g_{HP}=+0.16$ $g_{LP}=+0.07$



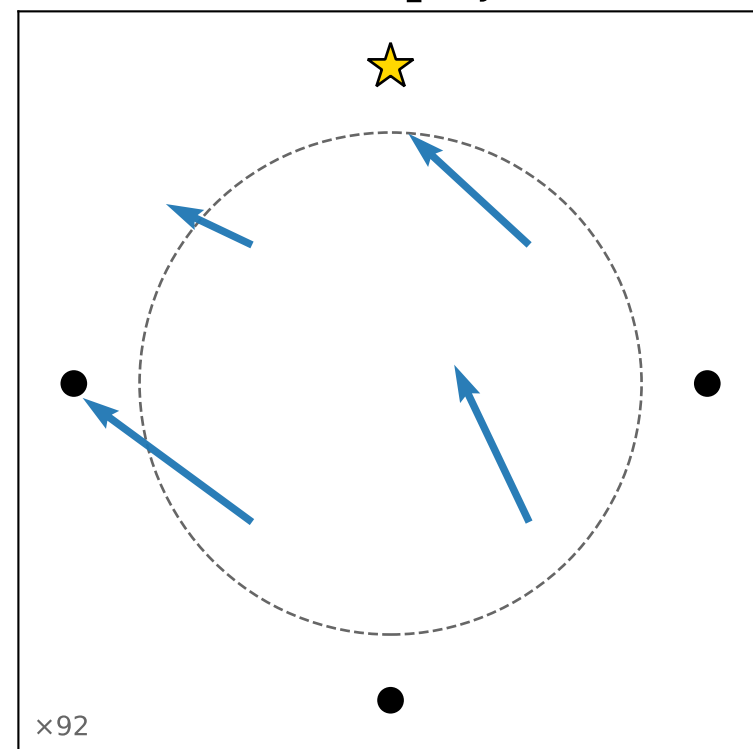
LO DATA $n_{\text{subj}} = 28$



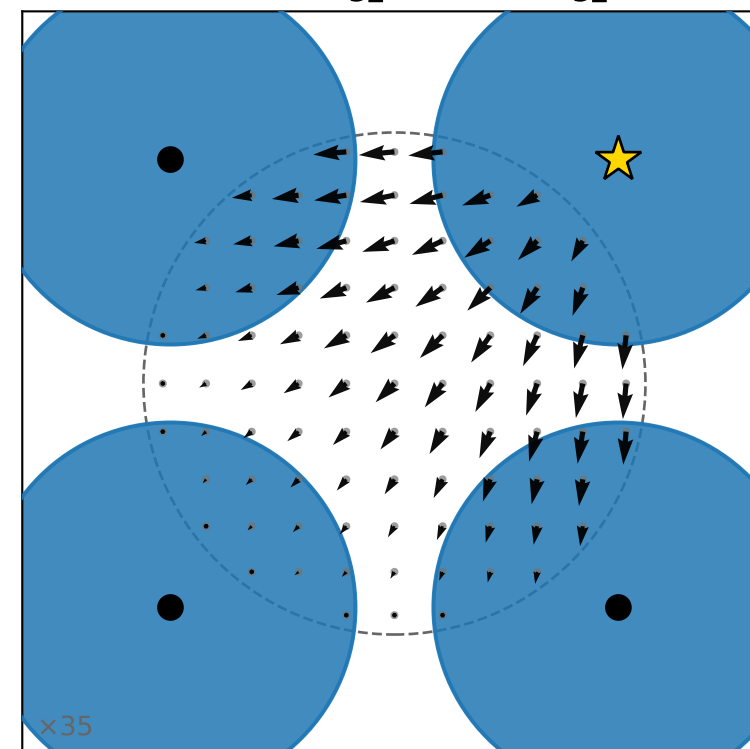
TO PRED $\sigma=2.57$ $g_{HP}=+0.15$ $g_{LP}=+0.05$



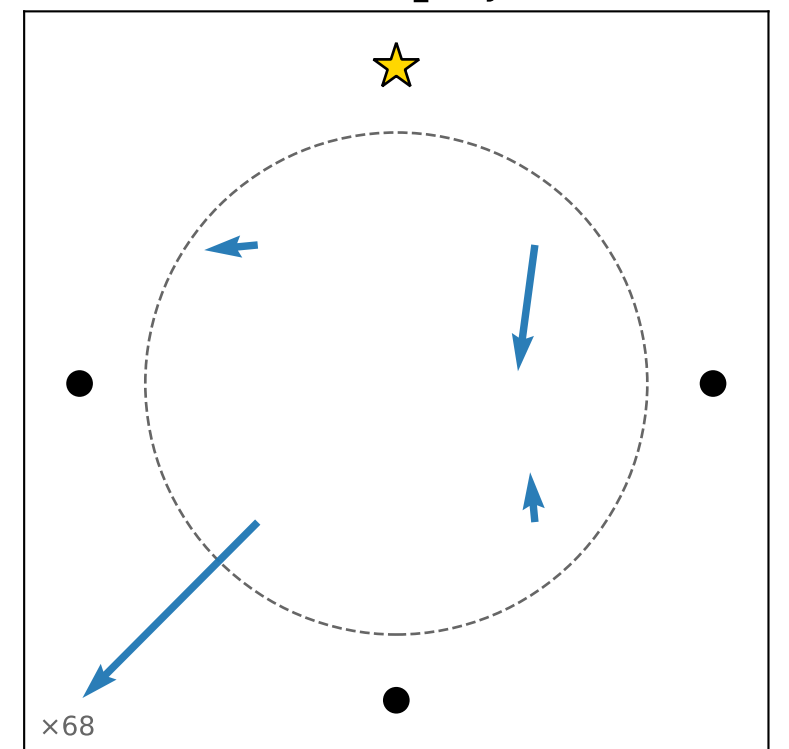
TO DATA $n_{\text{subj}} = 28$



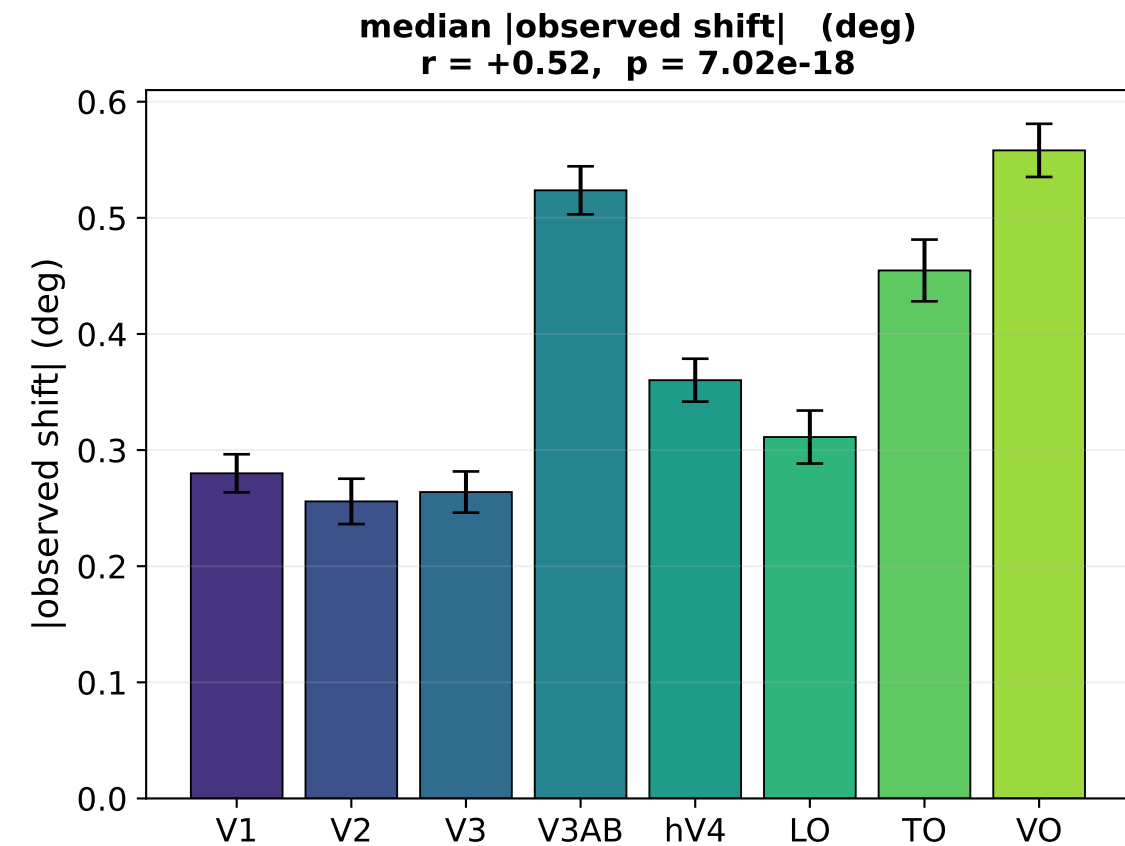
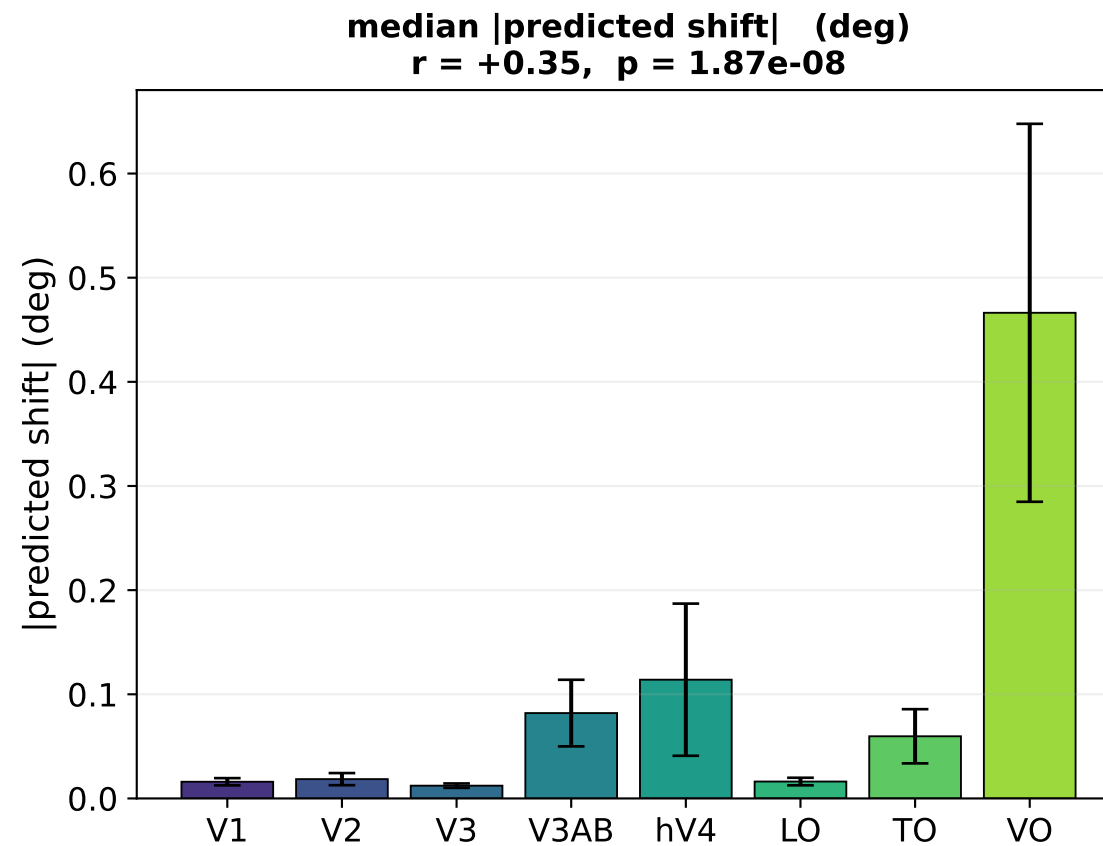
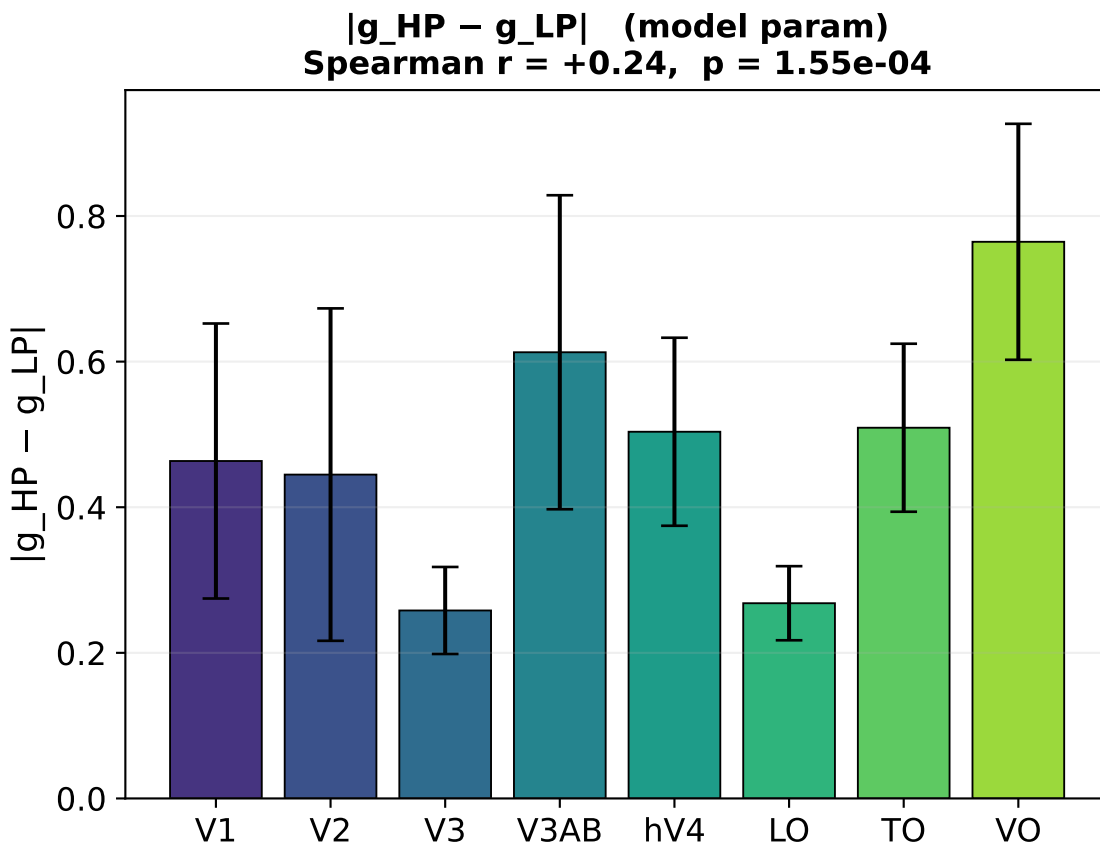
VO PRED $\sigma=2.34$ $g_{HP}=+0.62$ $g_{LP}=+0.72$



VO DATA $n_{\text{subj}} = 28$



(d'') Shifts grow up the visual hierarchy.
V1 → V2 → V3 → V3AB → hV4 → LO → TO → VO (mean across 30 subjects ± SEM).

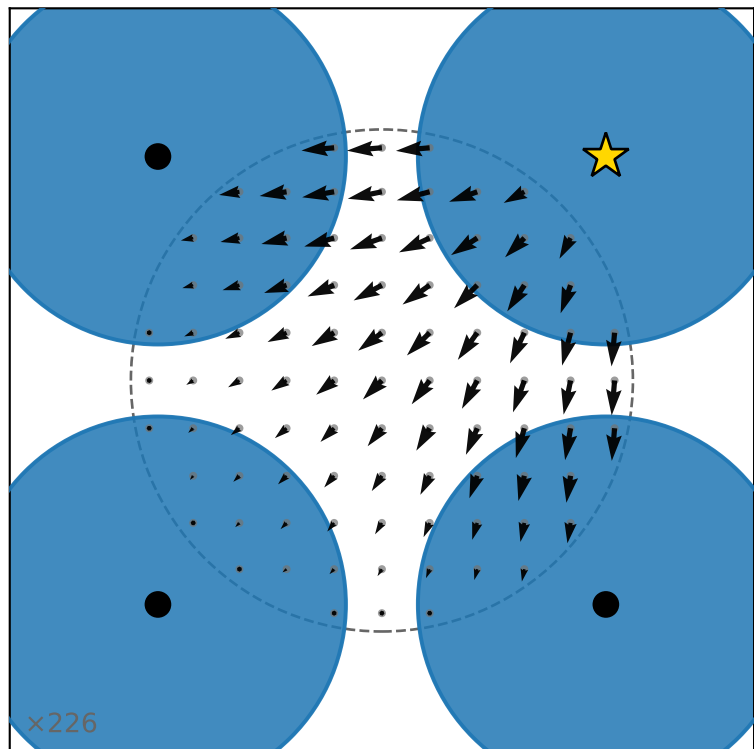


(d') Individual example subjects

sub-02, sub-25, sub-05 (top-3 by predicted-vs-observed r).

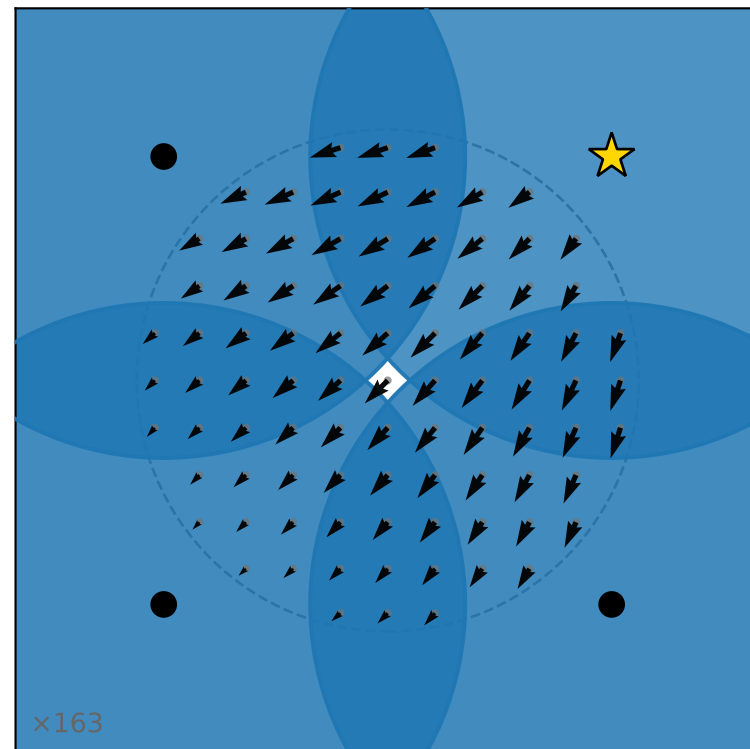
(d) Subject sub-02 — per-ROI predicted vs observed PRF-center shifts.
For each ROI: LEFT = model prediction, RIGHT = data.
Arrows: condition-specific shift (HP=★ – mean across 4 conditions for PRED;
mean-across-subjects of per-subject median shift for OBS).
Circles: each AF (size = σ_{AF} , shading $\propto |\text{gain}|$, red = suppression, blue = attraction).
Arrow $\times N$ annotation in each panel = auto-scale (85th-pctile arrow $\approx 70\%$ of a bin width, so PRED and OBS are comparable in shape).

V1 PRED $\sigma=2.37$ $g_{HP}=+0.23$ $g_{LP}=+0.24$



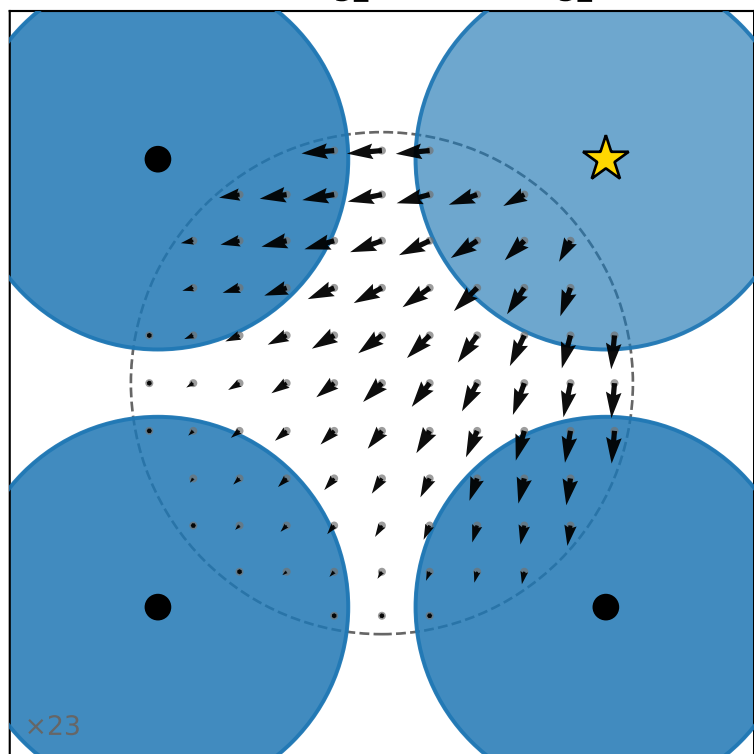
V1 DATA $n_{\text{subj}} = 0$

V2 PRED $\sigma=3.81$ $g_{HP}=+0.06$ $g_{LP}=+0.08$



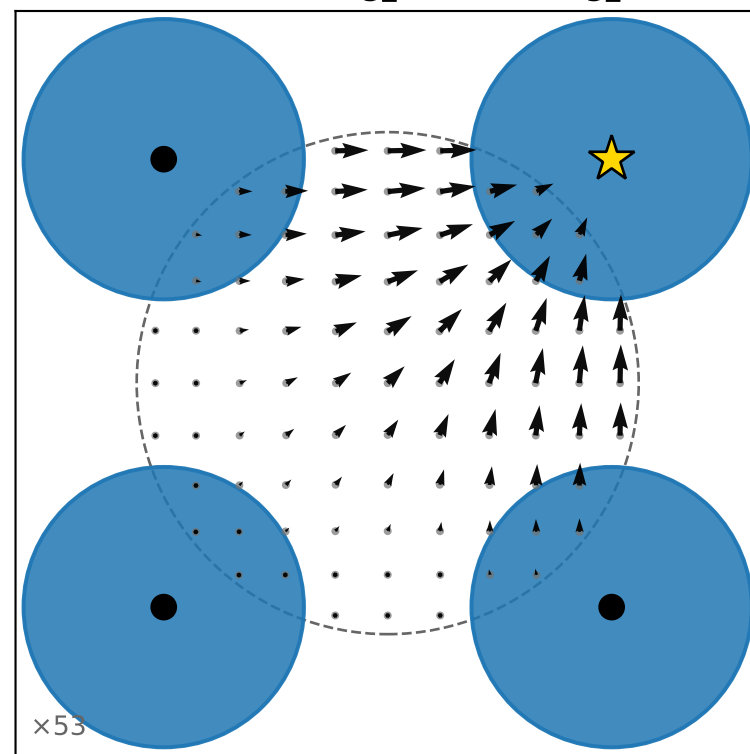
V2 DATA $n_{\text{subj}} = 0$

V3 PRED $\sigma=2.40$ $g_{HP}=+0.17$ $g_{LP}=+0.29$



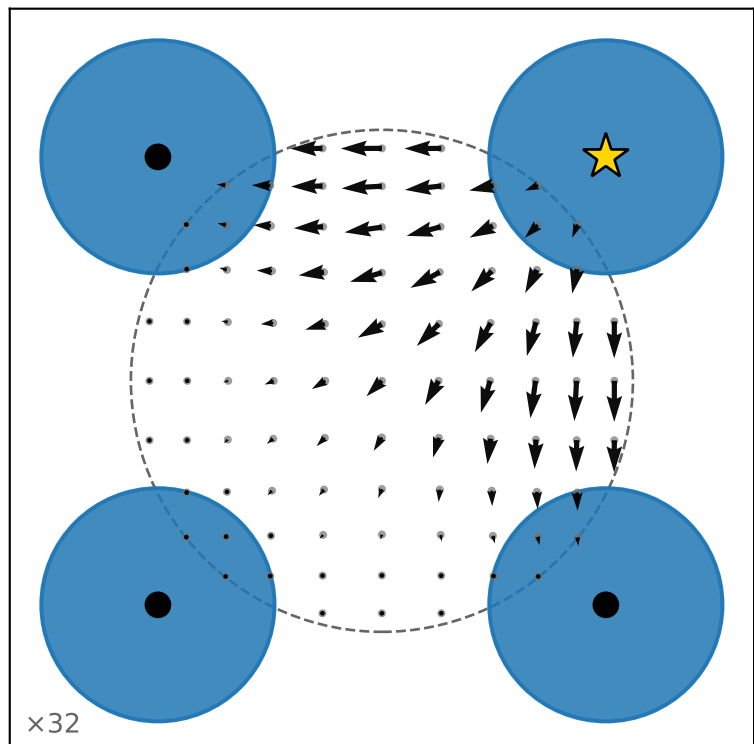
V3 DATA $n_{\text{subj}} = 0$

V3AB PRED $\sigma=1.77$ $g_{HP}=+1.84$ $g_{LP}=+1.75$



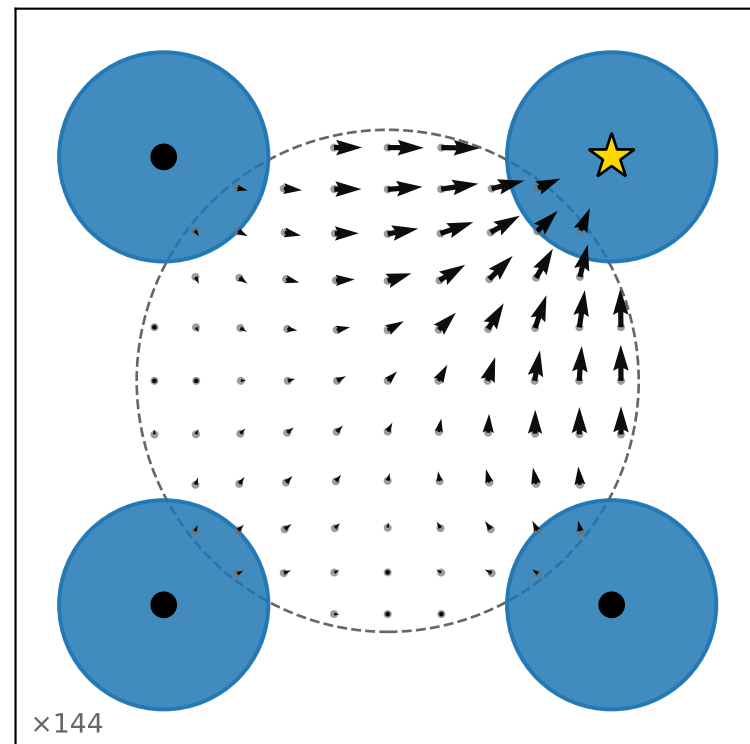
V3AB DATA $n_{\text{subj}} = 0$

hV4 PRED $\sigma=1.47$ $g_{HP}=+3.48$ $g_{LP}=+3.70$



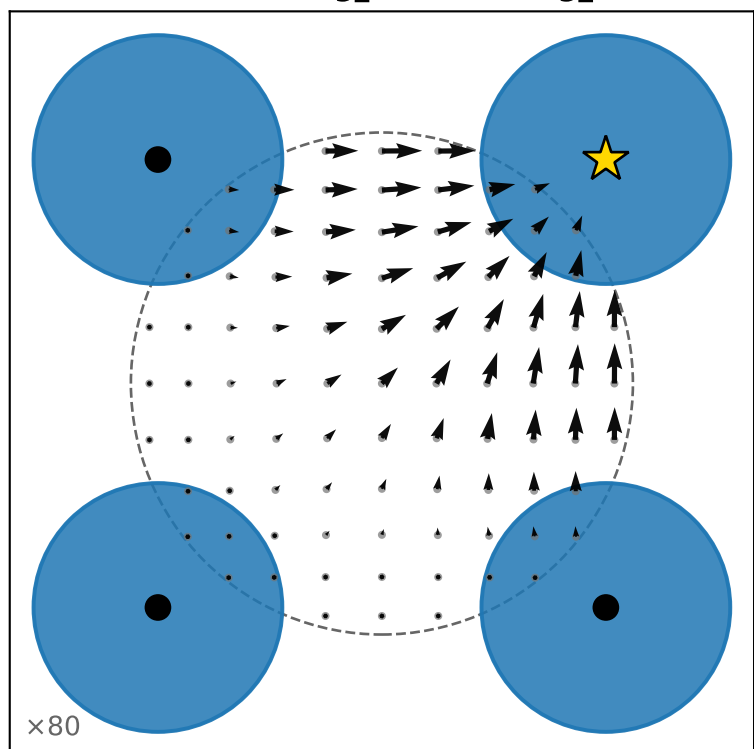
hV4 DATA $n_{\text{subj}} = 0$

LO PRED $\sigma=1.32$ $g_{HP}=+0.87$ $g_{LP}=+0.84$



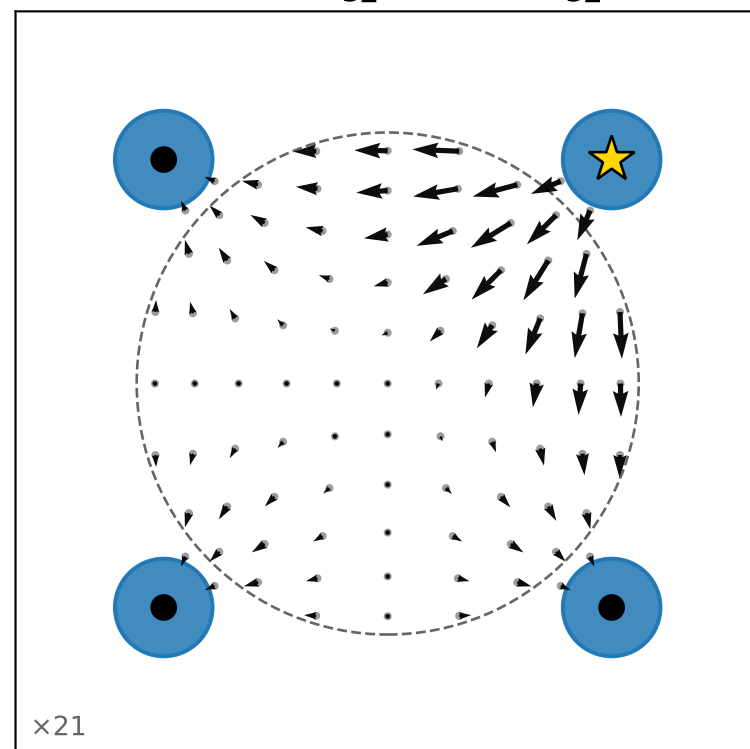
LO DATA $n_{\text{subj}} = 0$

TO PRED $\sigma=1.57$ $g_{HP}=+2.77$ $g_{LP}=+2.69$



TO DATA $n_{\text{subj}} = 0$

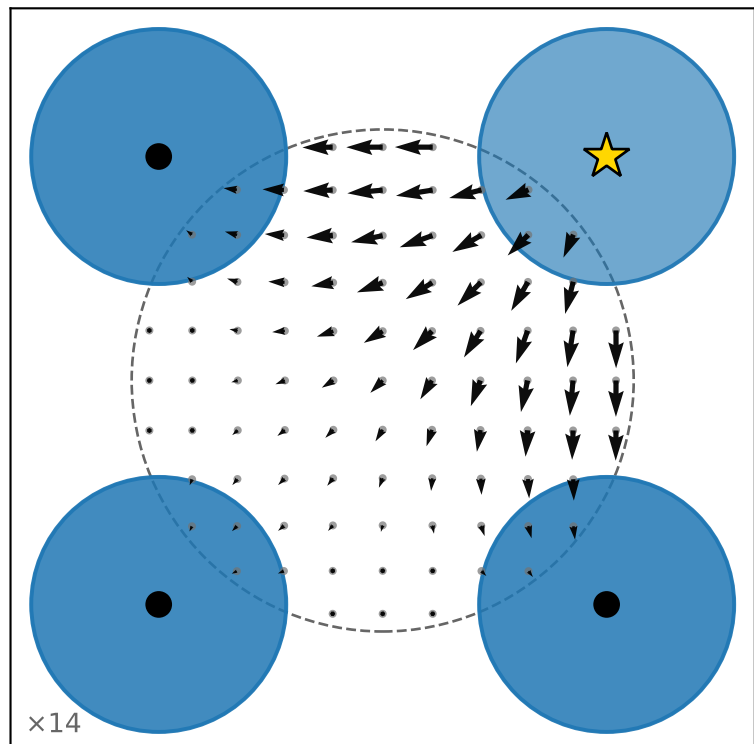
VO PRED $\sigma=0.62$ $g_{HP}=+5.57$ $g_{LP}=+6.33$



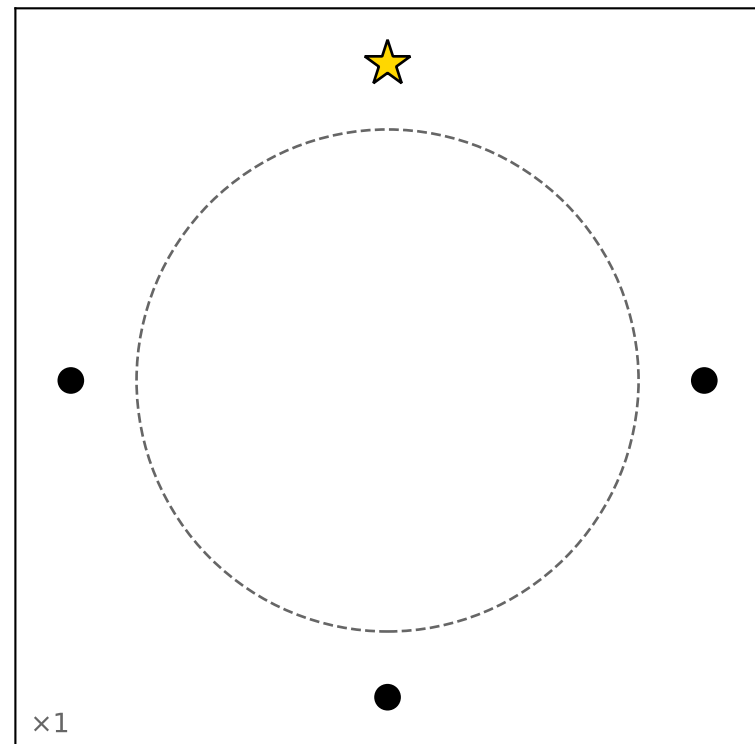
VO DATA $n_{\text{subj}} = 0$

(d) Subject sub-25 — per-ROI predicted vs observed PRF-center shifts.
For each ROI: LEFT = model prediction, RIGHT = data.
Arrows: condition-specific shift (HP=★ – mean across 4 conditions for PRED;
mean-across-subjects of per-subject median shift for OBS).
Circles: each AF (size = σ_{AF} , shading $\propto |\text{gain}|$, red = suppression, blue = attraction).
Arrow $\times N$ annotation in each panel = auto-scale (85th-pctile arrow $\approx 70\%$ of a bin width, so PRED and OBS are comparable in shape).

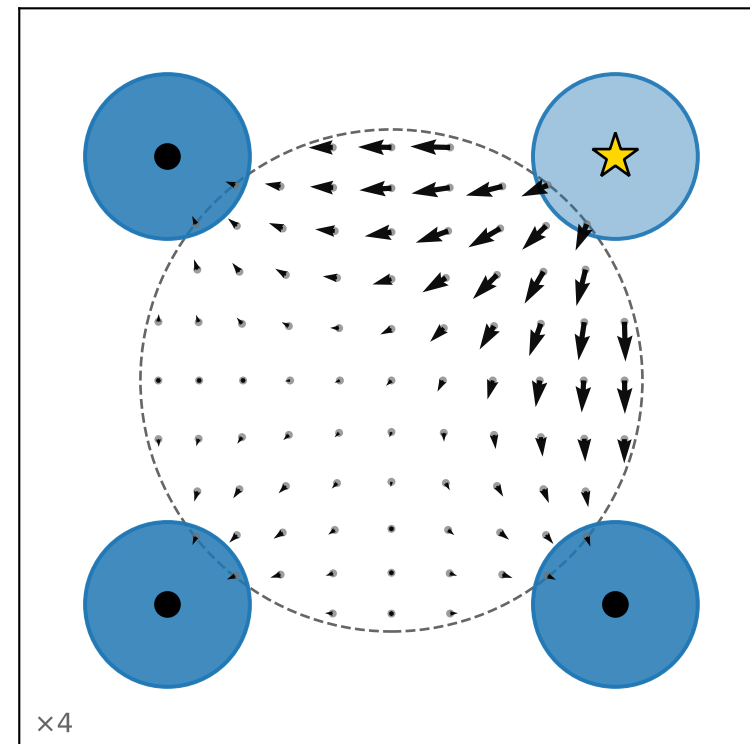
V1 PRED $\sigma=1.61$ $g_{HP}=+0.27$ $g_{LP}=+0.48$



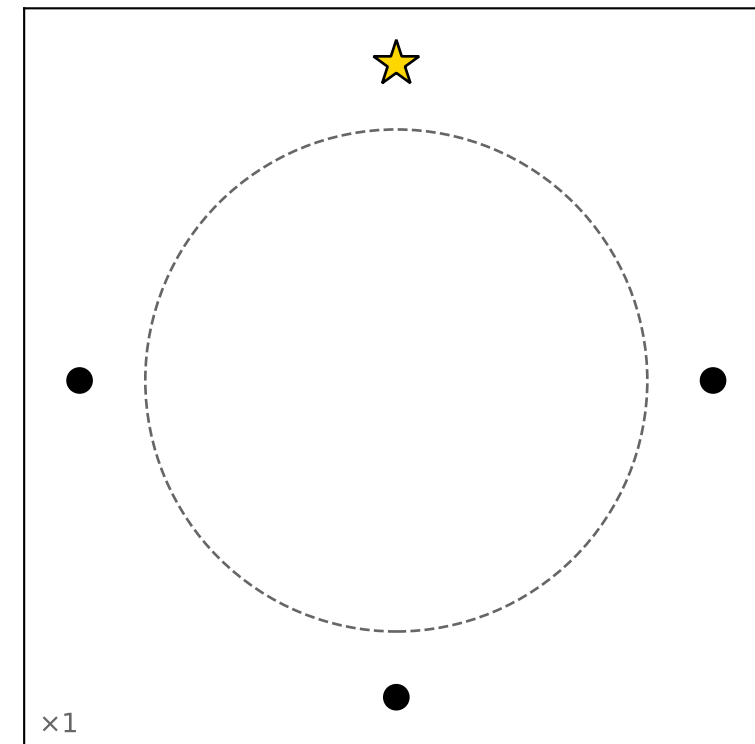
V1 DATA $n_{\text{subj}} = 1$



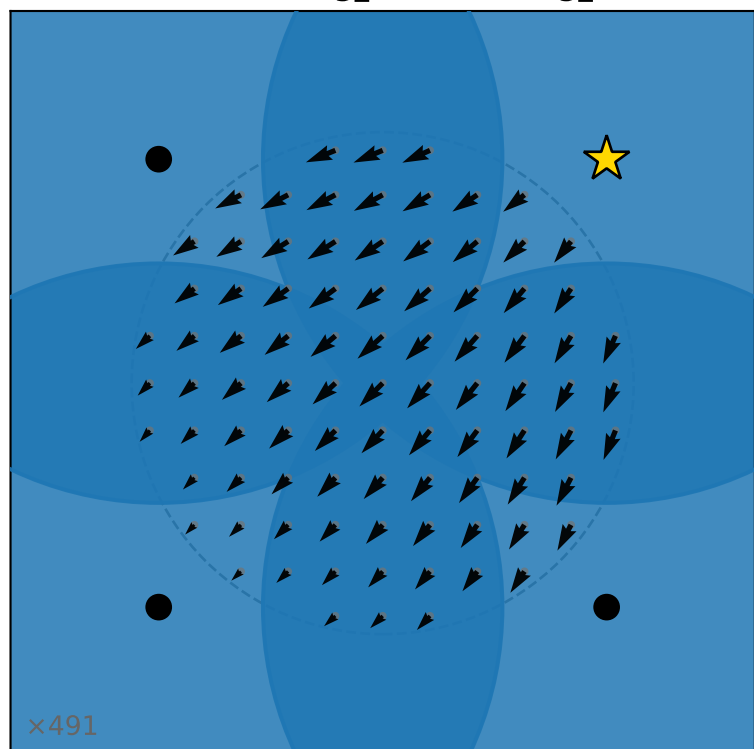
V2 PRED $\sigma=1.04$ $g_{HP}=+0.61$ $g_{LP}=+1.92$



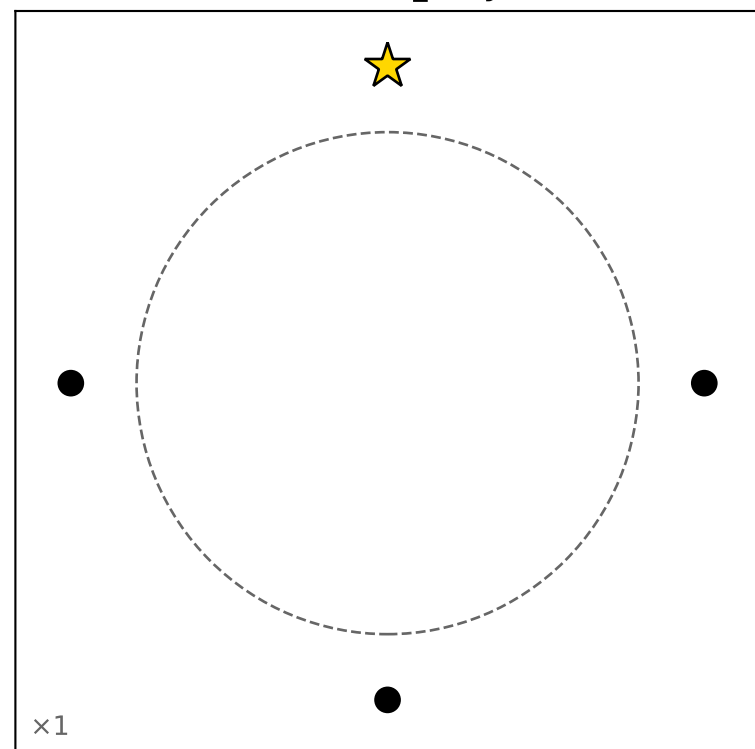
V2 DATA $n_{\text{subj}} = 1$



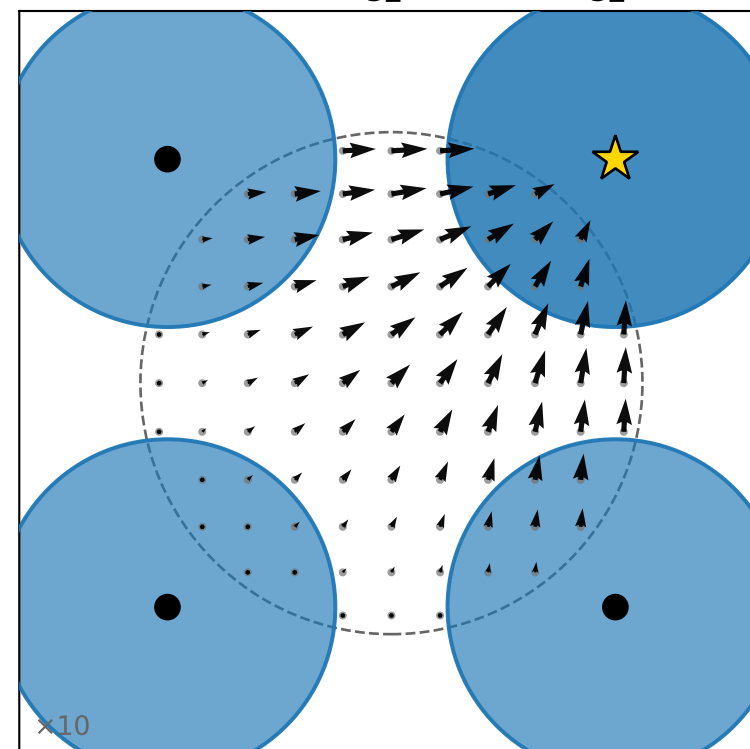
V3 PRED $\sigma=4.34$ $g_{HP}=+0.07$ $g_{LP}=+0.08$



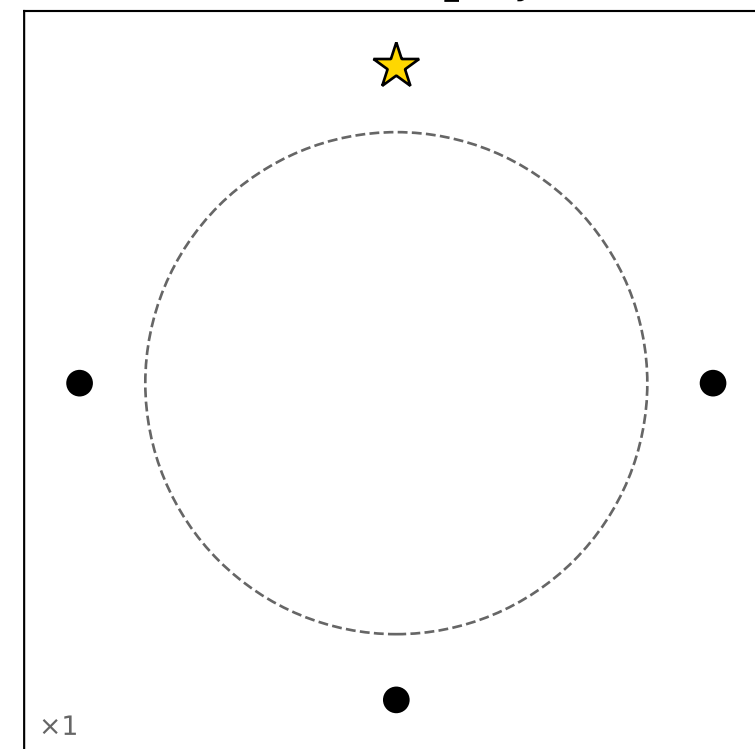
V3 DATA $n_{\text{subj}} = 1$



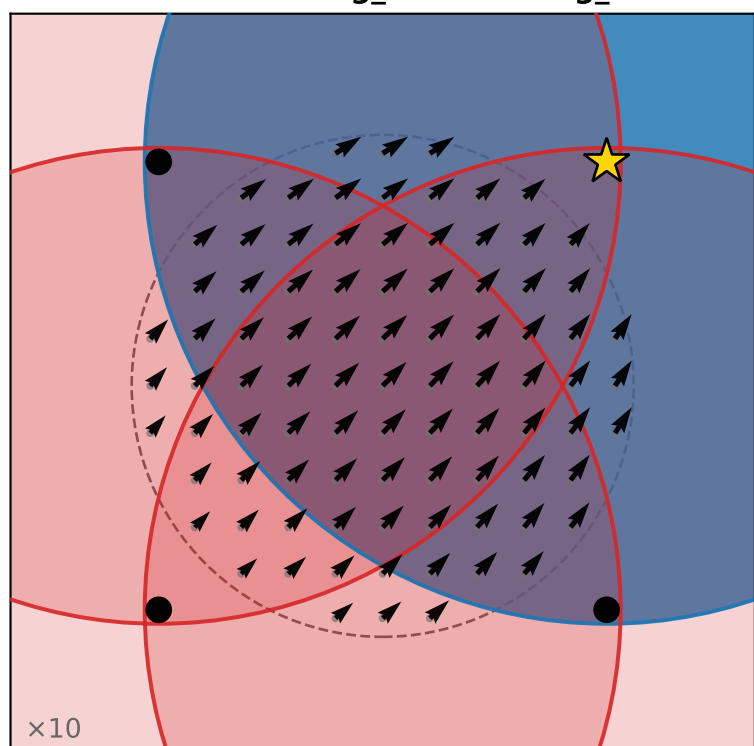
V3AB PRED $\sigma=2.12$ $g_{HP}=+0.86$ $g_{LP}=+0.51$



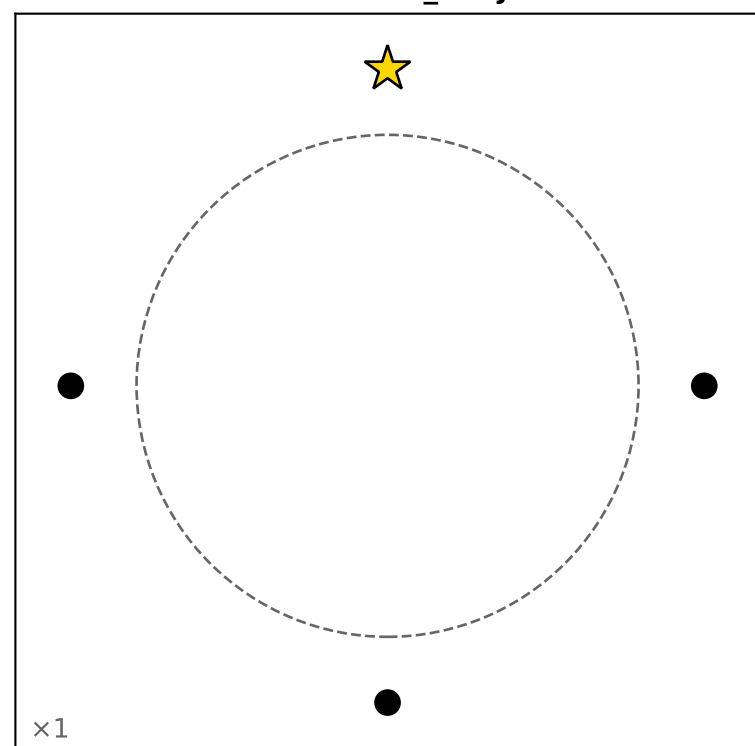
V3AB DATA $n_{\text{subj}} = 1$



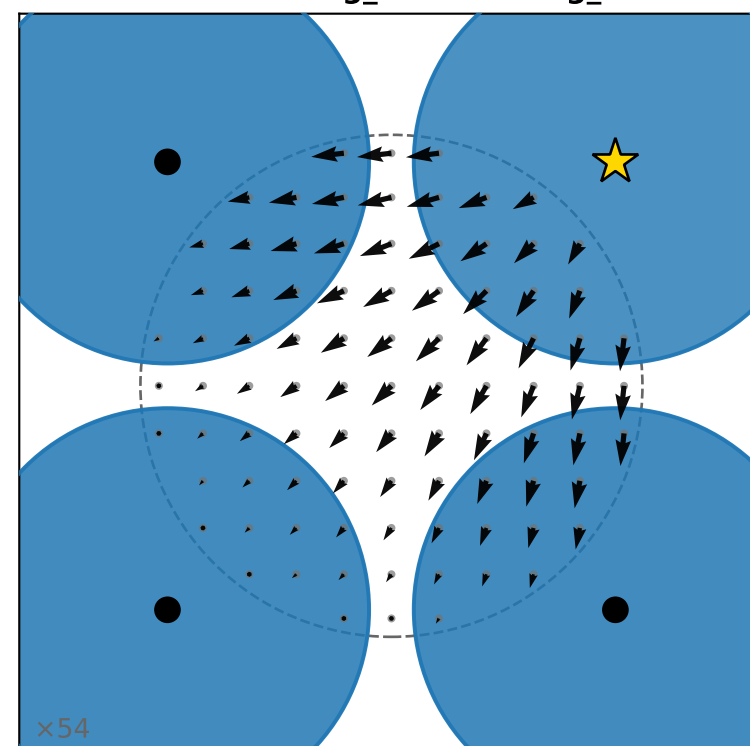
hV4 PRED $\sigma=5.83$ $g_{HP}=+0.65$ $g_{LP}=-0.05$



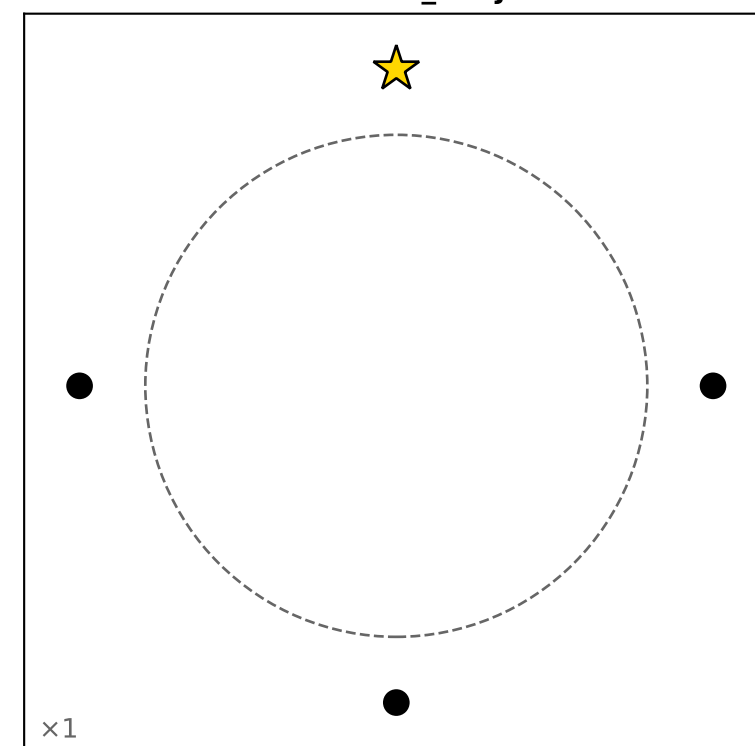
hV4 DATA $n_{\text{subj}} = 1$



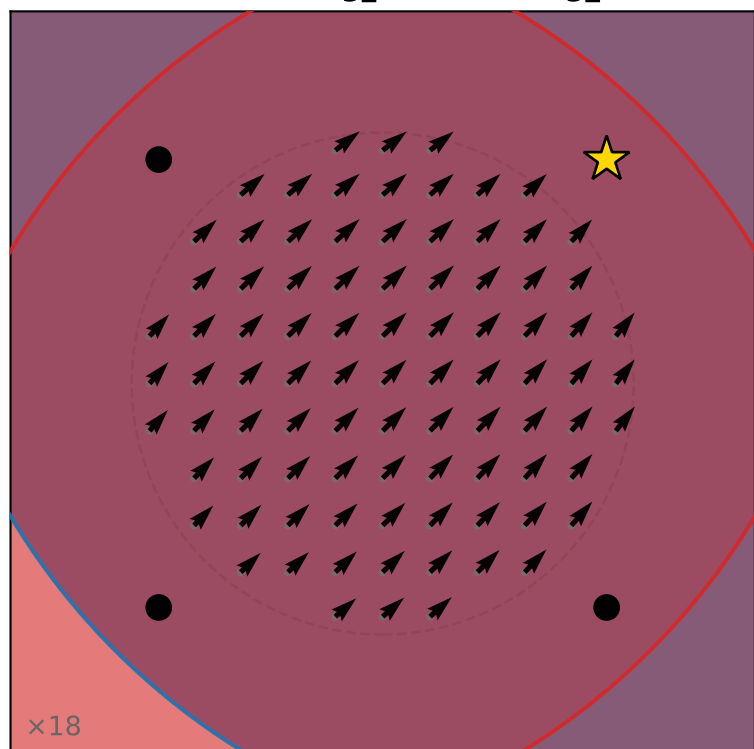
LO PRED $\sigma=2.55$ $g_{HP}=+0.17$ $g_{LP}=+0.22$



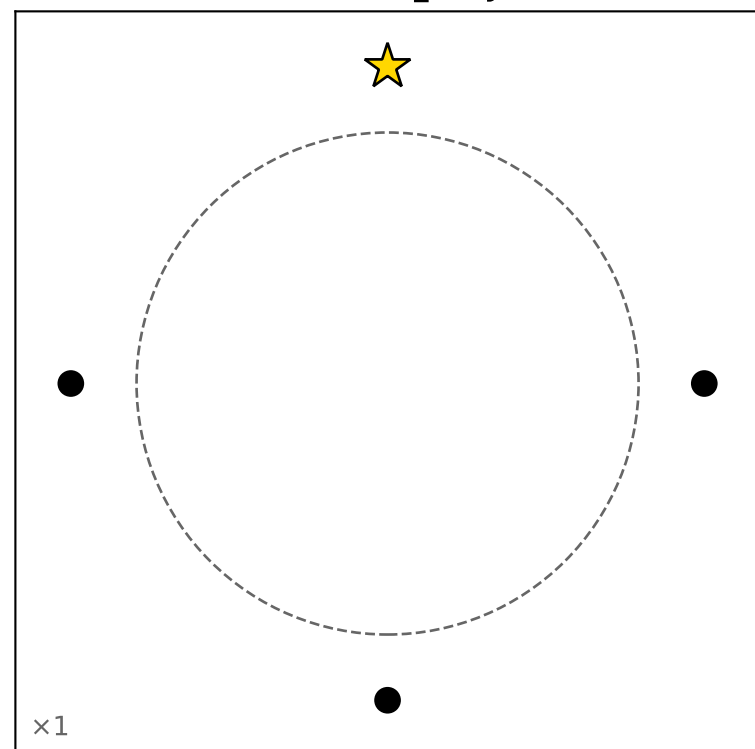
LO DATA $n_{\text{subj}} = 1$



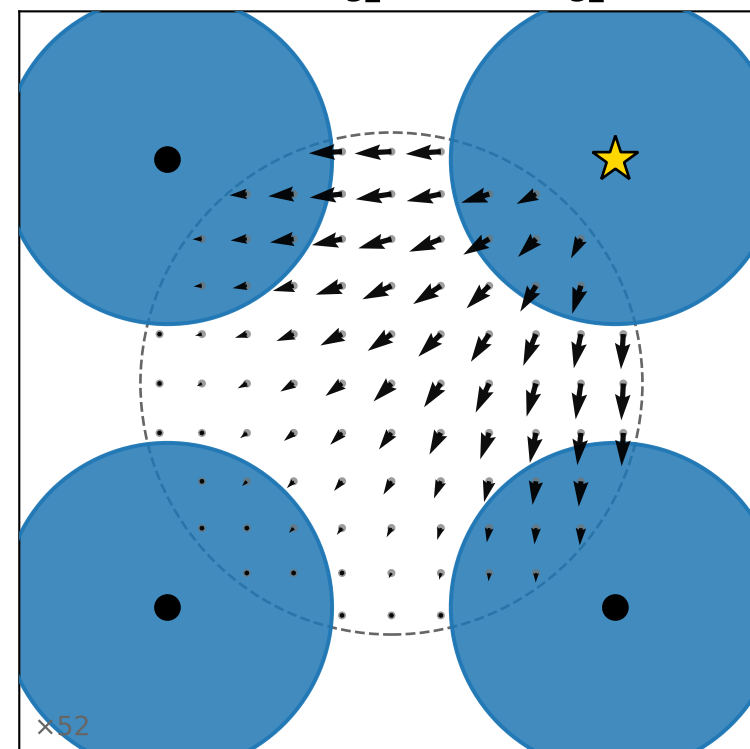
TO PRED $\sigma=8.77$ $g_{HP}=+0.60$ $g_{LP}=-0.09$



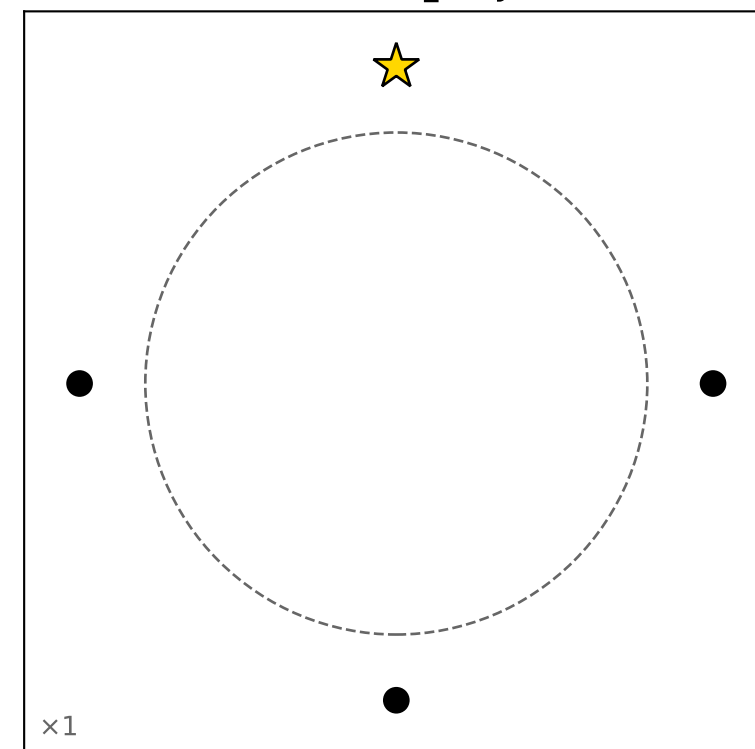
TO DATA $n_{\text{subj}} = 1$



VO PRED $\sigma=2.08$ $g_{HP}=+1.02$ $g_{LP}=+1.10$

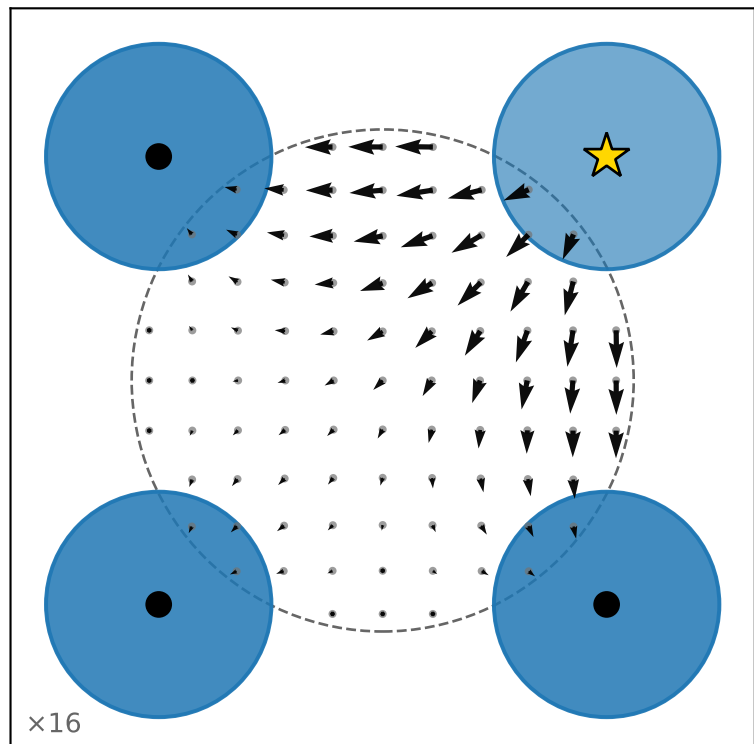


VO DATA $n_{\text{subj}} = 1$

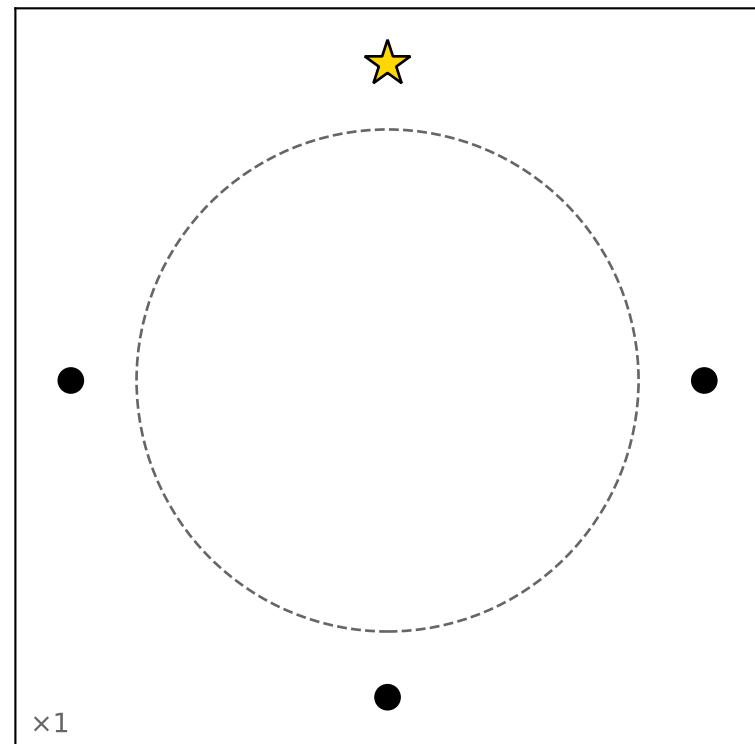


(d) Subject sub-05 — per-ROI predicted vs observed PRF-center shifts.
For each ROI: LEFT = model prediction, RIGHT = data.
Arrows: condition-specific shift (HP=★ – mean across 4 conditions for PRED;
mean-across-subjects of per-subject median shift for OBS).
Circles: each AF (size = σ_{AF} , shading $\propto |\text{gain}|$, red = suppression, blue = attraction).
Arrow $\times N$ annotation in each panel = auto-scale (85th-pctile arrow $\approx 70\%$ of a bin width, so PRED and OBS are comparable in shape).

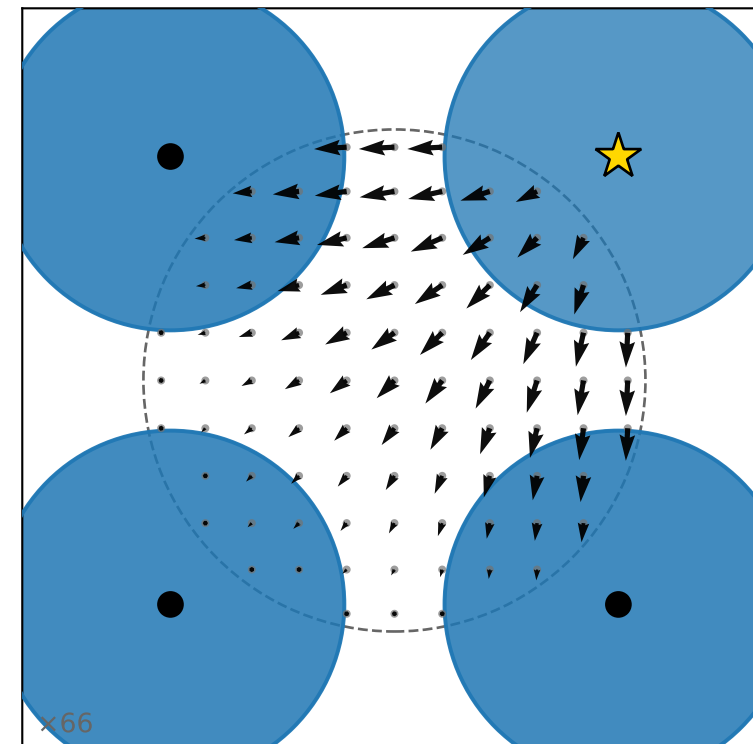
V1 PRED $\sigma=1.42$ $g_{HP}=+0.22$ $g_{LP}=+0.40$



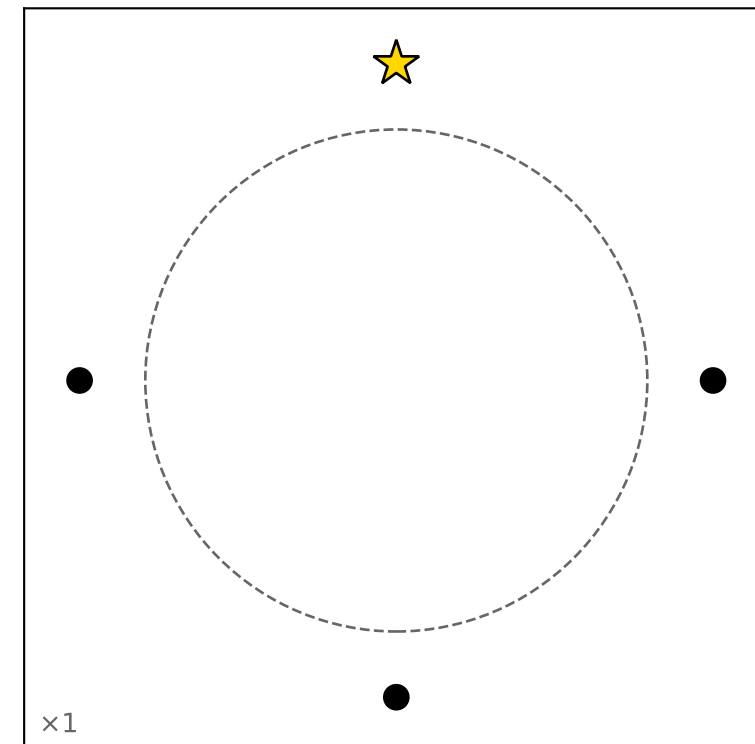
V1 DATA $n_{\text{subj}} = 1$



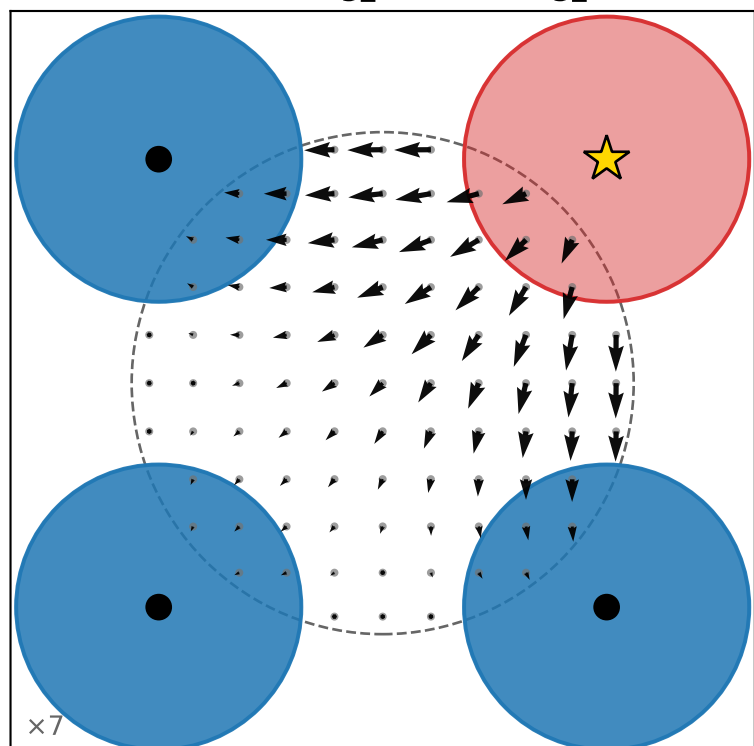
V2 PRED $\sigma=2.20$ $g_{HP}=+0.09$ $g_{LP}=+0.13$



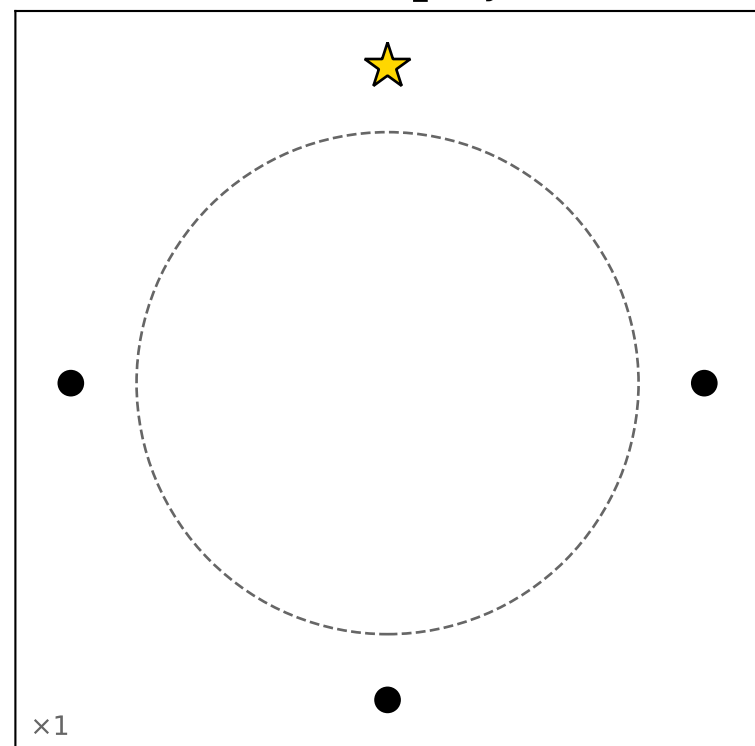
V2 DATA $n_{\text{subj}} = 1$



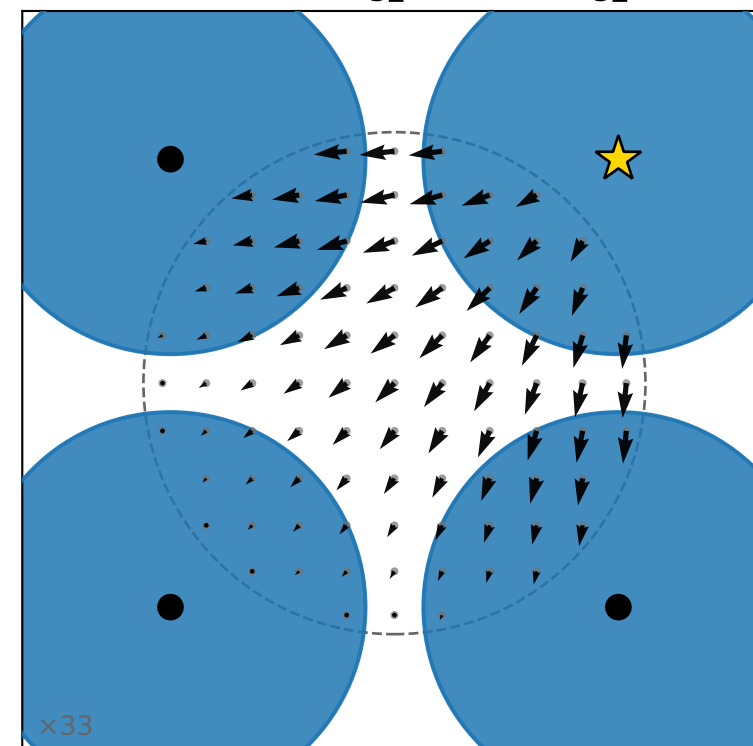
V3 PRED $\sigma=1.80$ $g_{HP}=-0.09$ $g_{LP}=+0.25$



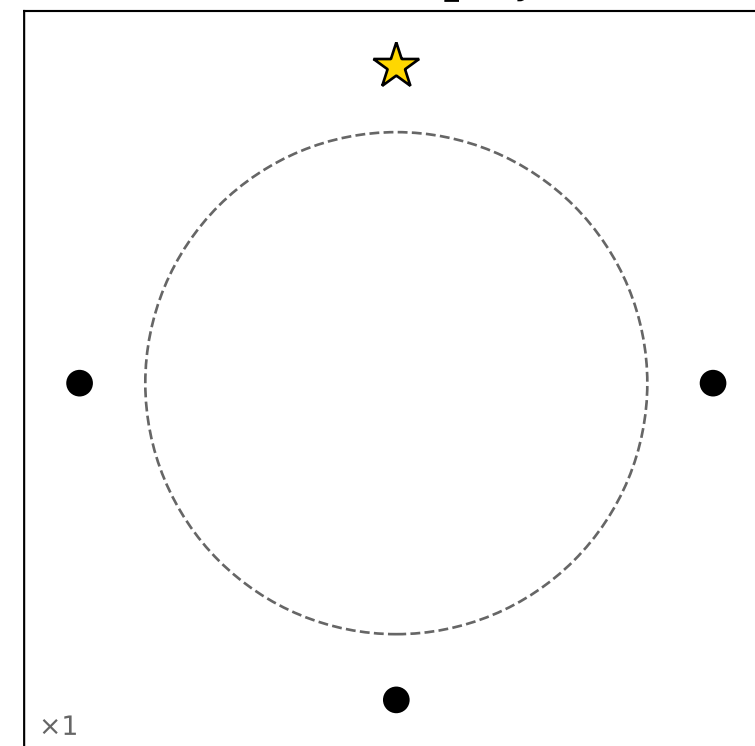
V3 DATA $n_{\text{subj}} = 1$



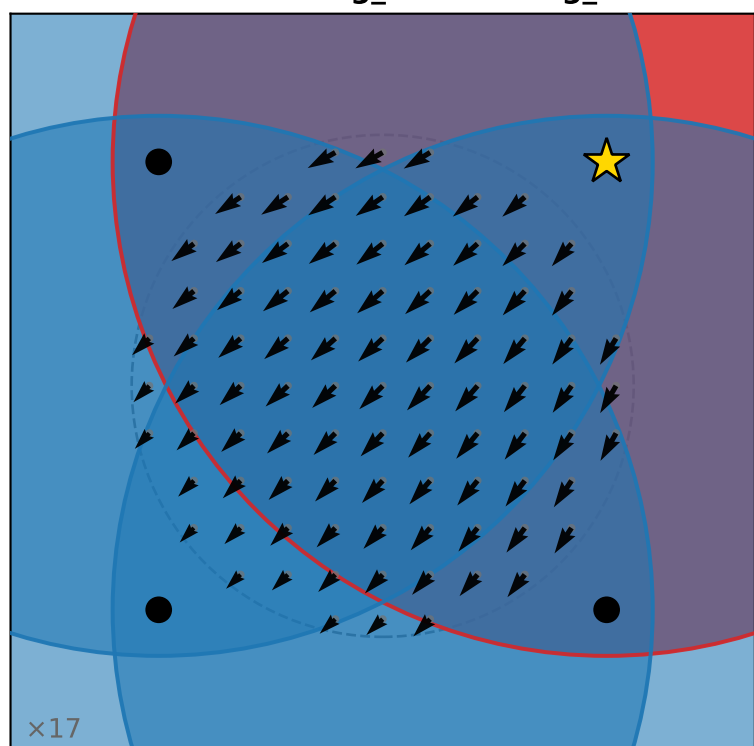
V3AB PRED $\sigma=2.46$ $g_{HP}=+0.37$ $g_{LP}=+0.47$



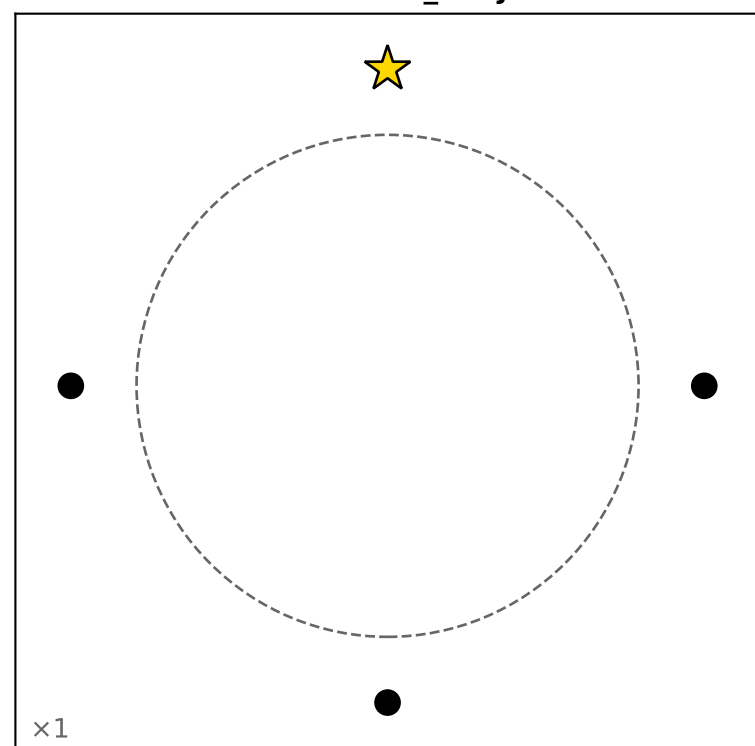
V3AB DATA $n_{\text{subj}} = 1$



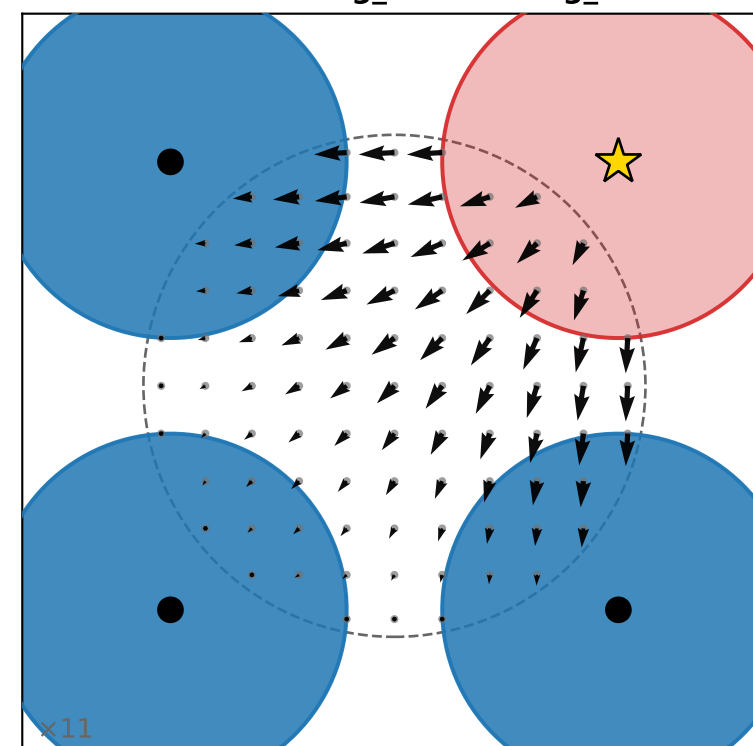
hV4 PRED $\sigma=6.24$ $g_{HP}=-0.22$ $g_{LP}=+0.11$



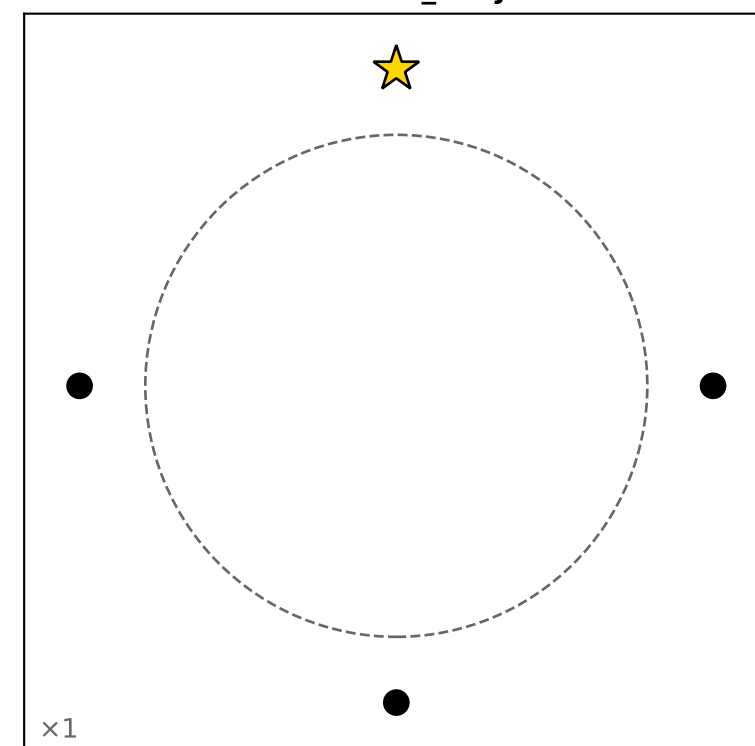
hV4 DATA $n_{\text{subj}} = 1$



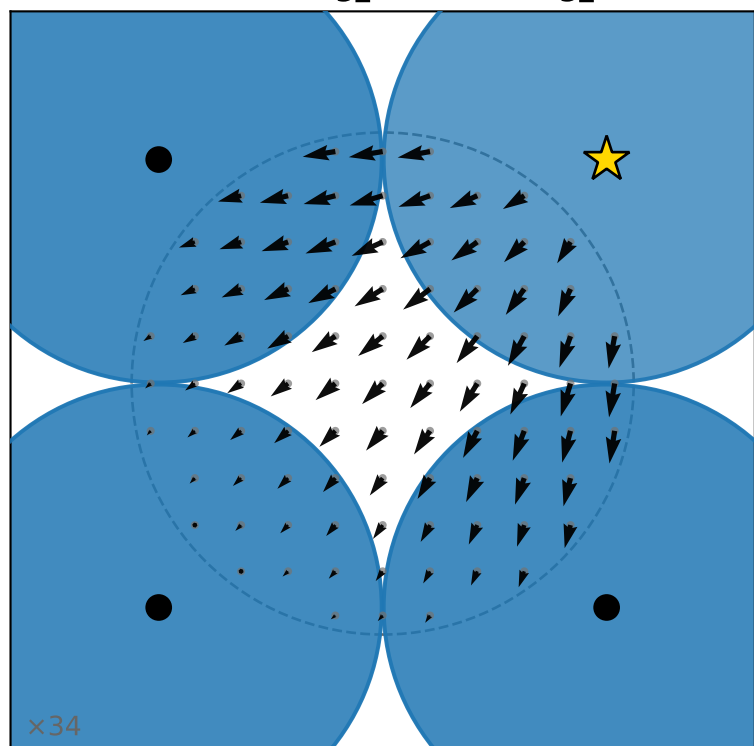
LO PRED $\sigma=2.22$ $g_{HP}=-0.03$ $g_{LP}=+0.17$



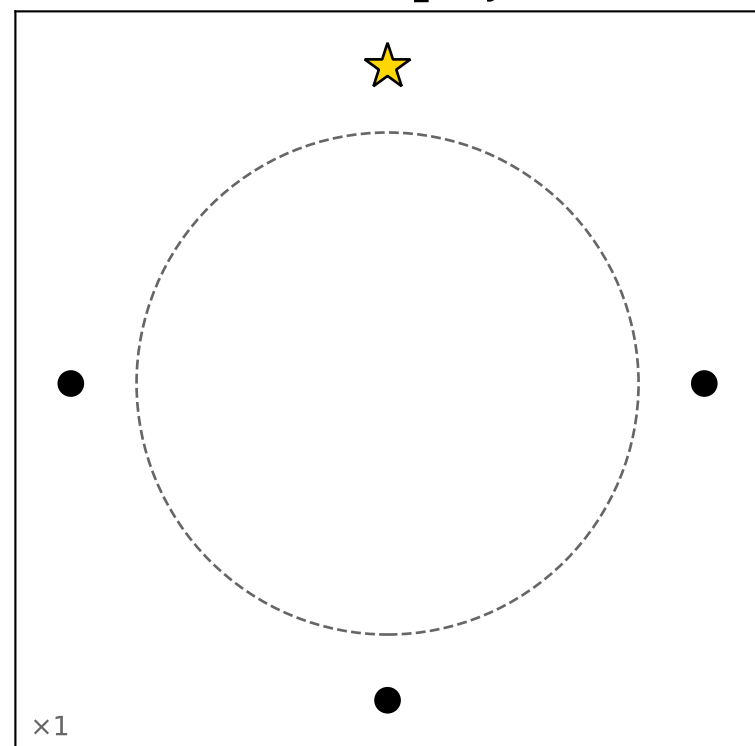
LO DATA $n_{\text{subj}} = 1$



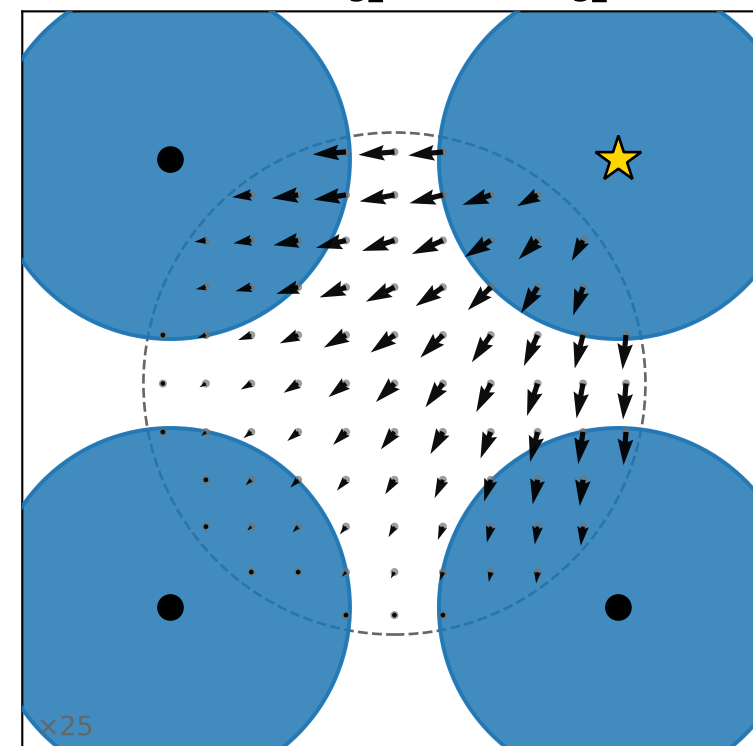
TO PRED $\sigma=2.82$ $g_{HP}=+0.20$ $g_{LP}=+0.29$



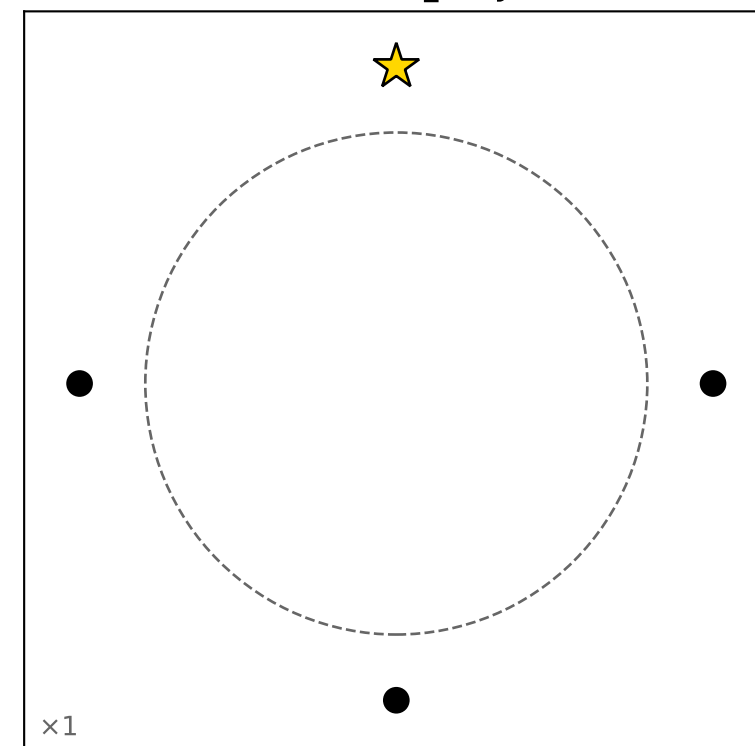
TO DATA $n_{\text{subj}} = 1$



VO PRED $\sigma=2.27$ $g_{HP}=+0.75$ $g_{LP}=+0.91$



VO DATA $n_{\text{subj}} = 1$

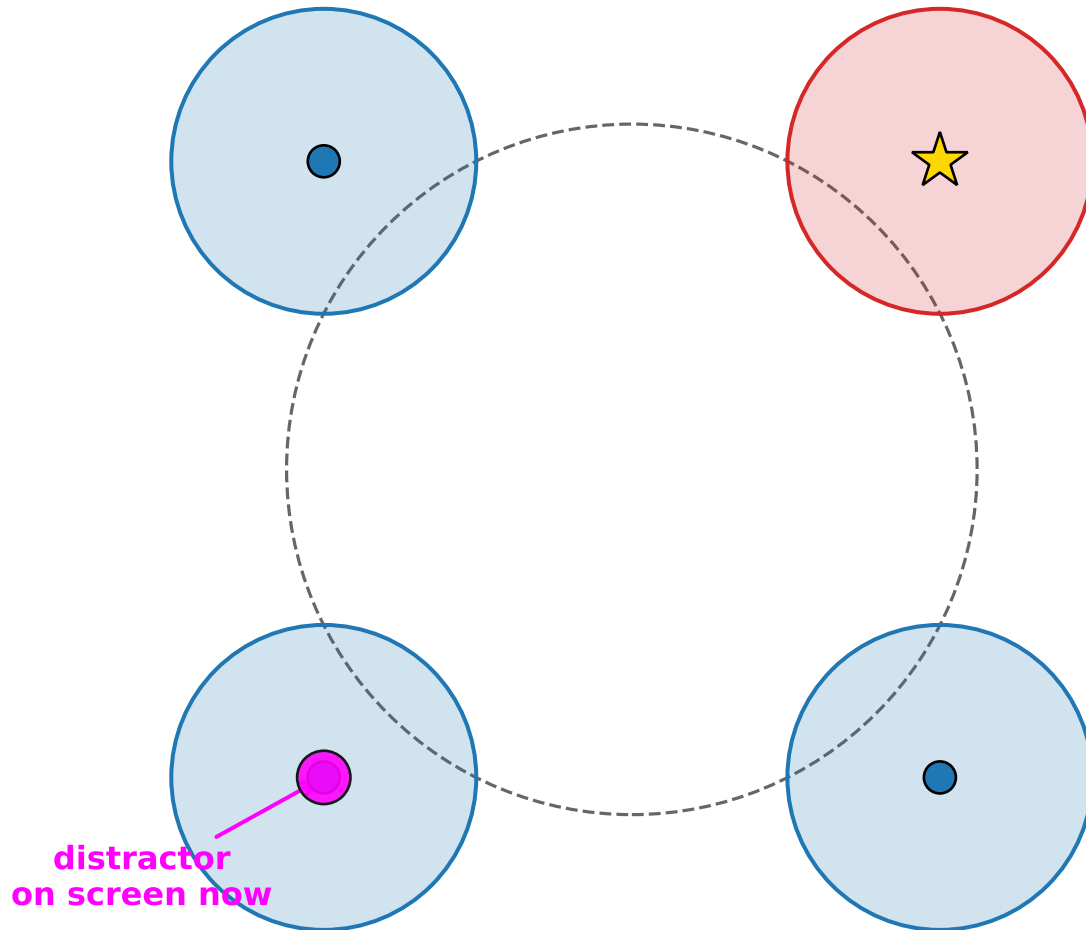


ic AF — distractor pulses on top of the sustained p

5 shared parameters per ROI; same per-voxel Gaussian kernel.

(e) Dynamic AF — adds a per-TR distractor pulse

sustained AF at all 4 ring locations
+ TRANSIENT AF at the location currently
showing a distractor (on/off per TR)



Five shared parameters per ROI

σ_{AF} AF size (shared sustained $\leftrightarrow$ dynamic)

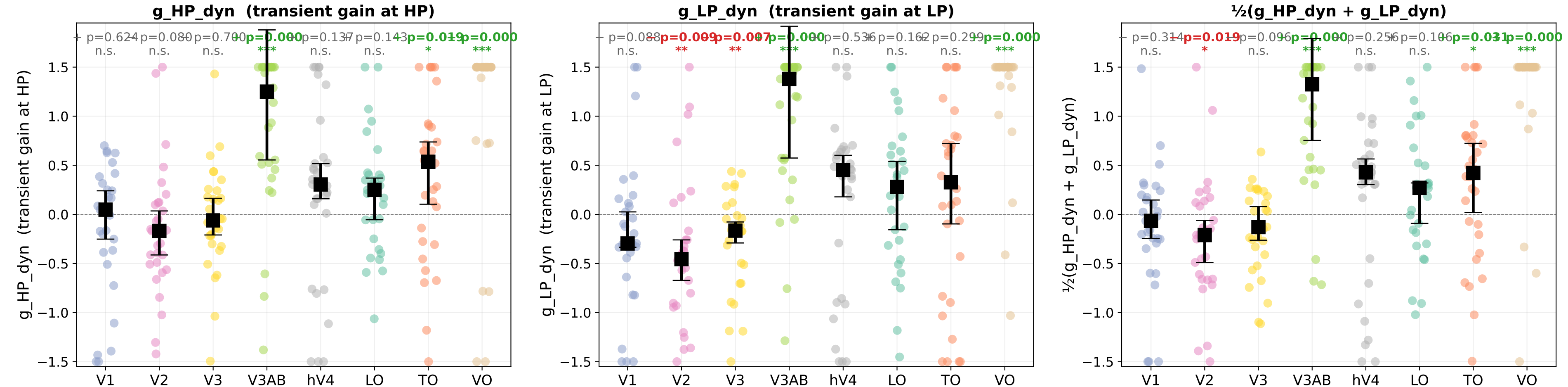
g_{HP} sustained gain at HP
 g_{LP} sustained gain at LP

g_{HP_dyn} transient gain when distractor at HP
 g_{LP_dyn} transient gain when distractor at LP

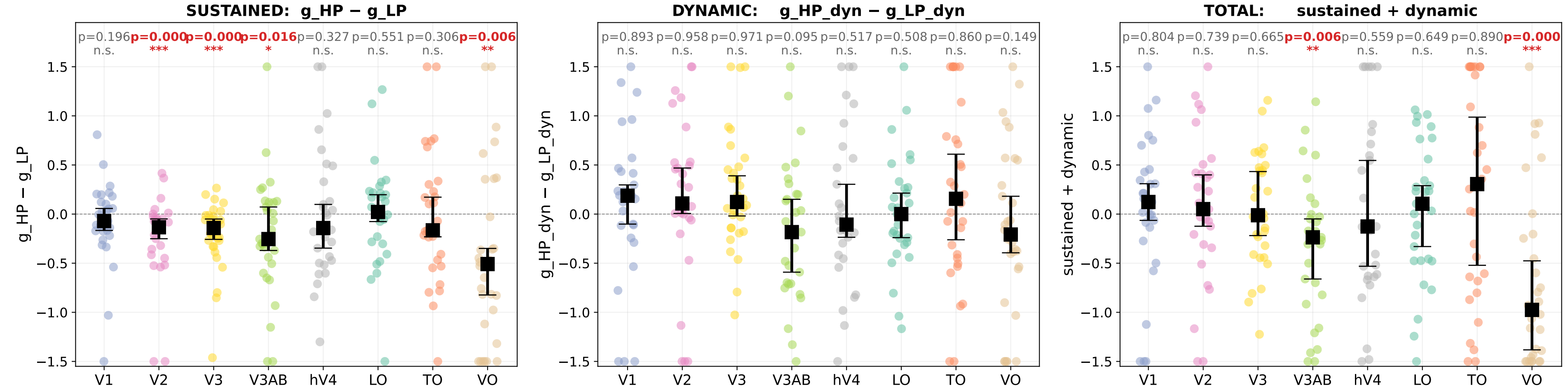
$$M(g, t) = 1 + g_{HP} \cdot A_{HP_C}(g) + g_{LP} \cdot \sum A_{LP}(g) + g_{HP_dyn} \cdot d_{HP}(t) \cdot A_{HP_C}(g) + g_{LP_dyn} \cdot \sum d_{LP}(t) \cdot A_{LP}(g)$$

$d_{\square}(t) \in [0, 1]$: fraction of TR with distractor on screen at ring location $\square$.

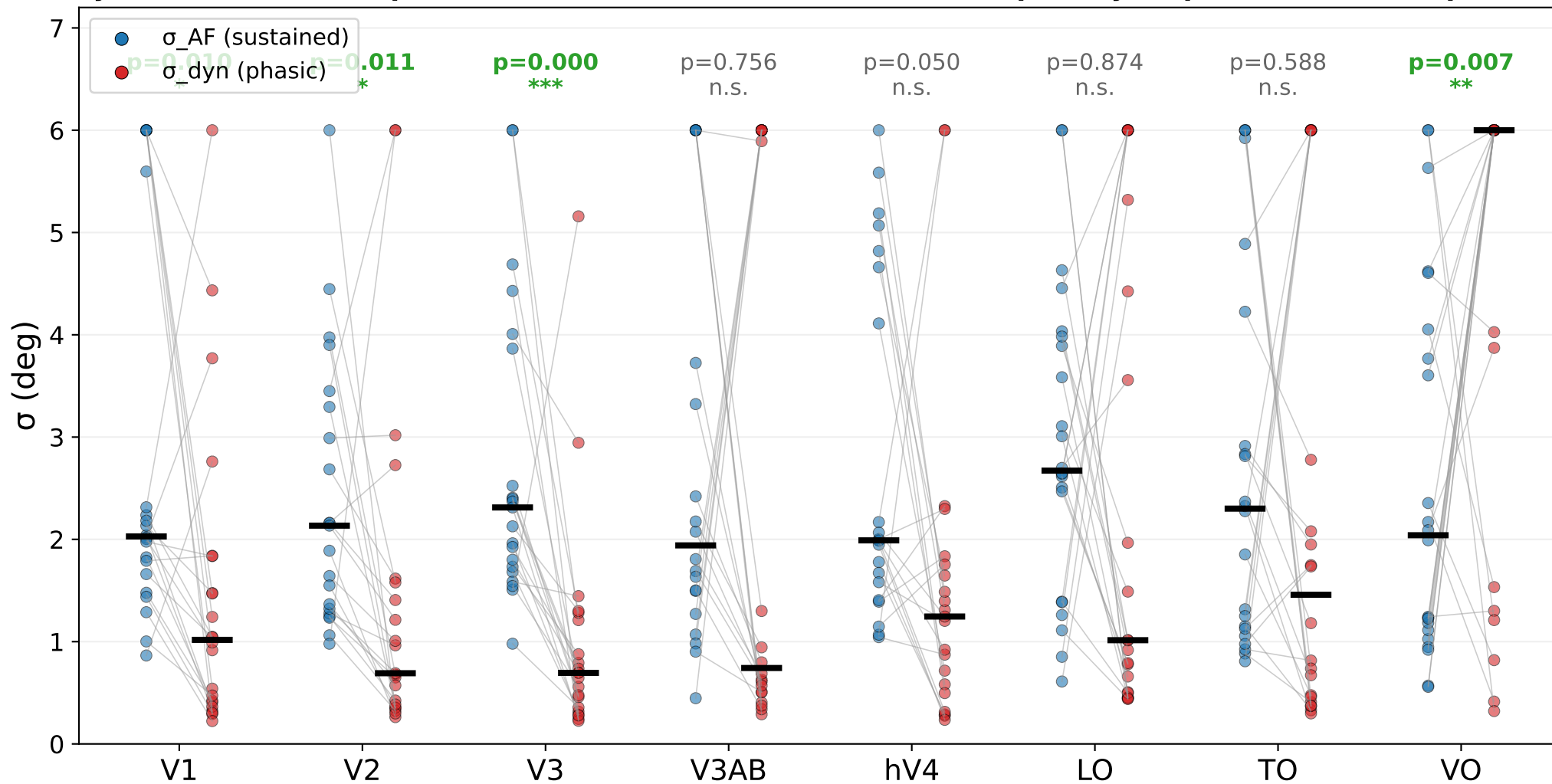
(e) Is there transient attention at all? g_{HP_dyn} and g_{LP_dyn} vs zero per ROI.
GREEN = significant CAPTURE (attraction). RED = significant SUPPRESSION.



(e) HP–LP differential: sustained, dynamic, and total per ROI.
Wilcoxon one-sided H_1 : HP < LP. V2/V3 cancel — dynamic offsets sustained. V3AB/VO add.



σ_{AF} (sustained, blue) vs σ_{dyn} (phasic distractor, red) — v3 model, per-fit paired comparison
In early visual cortex, the phasic AF is narrower than the sustained priority map → two distinct spatial scales.

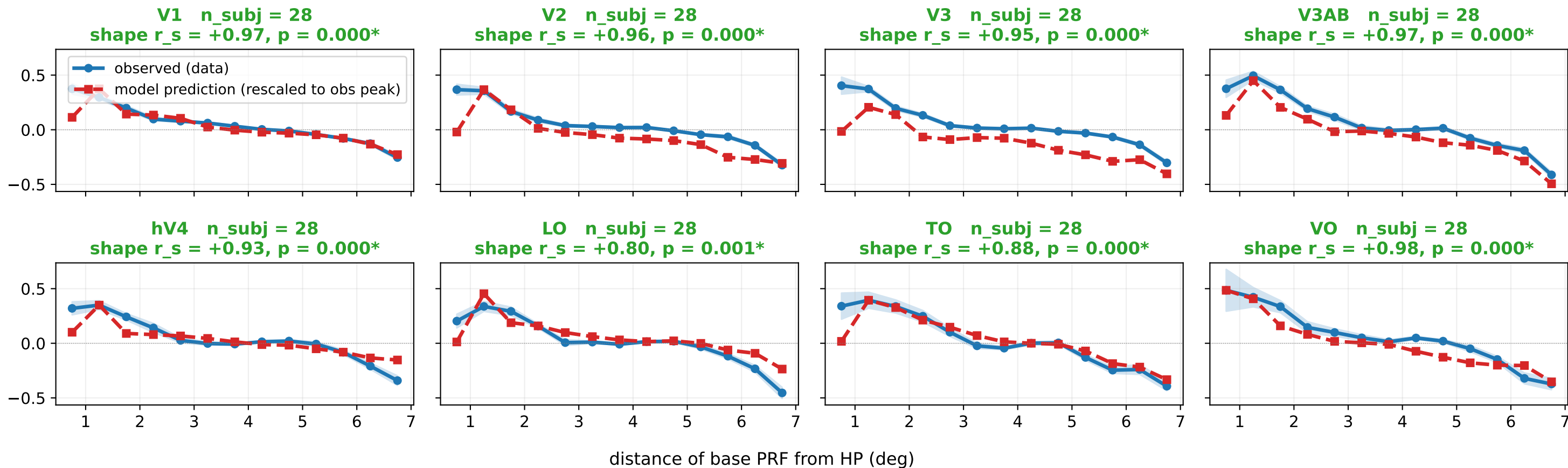


(g) Observed vs predicted shifts

Per-ROI projection-vs-distance from HP. Solid = data, dashed = model.

(g) Qualitative pattern test: model SHAPE vs data SHAPE.
Per ROI: projection on away-from-HP axis vs distance from HP.
Predicted curve rescaled to match observed peak — magnitude is confounded by Jensen & kernel mismatch, but the SHAPE (peak location, decay) is the test.

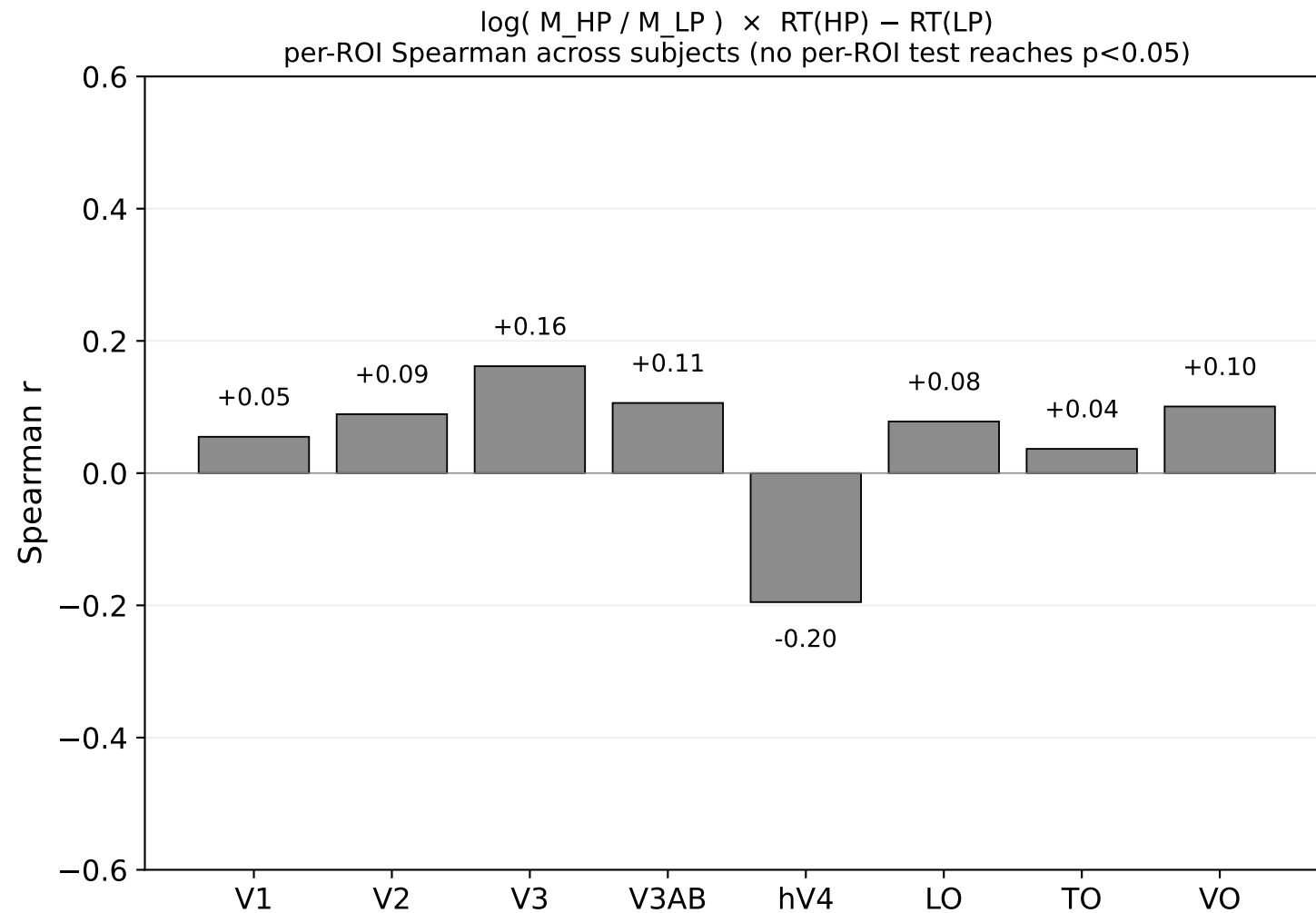
projection on away-from-HP axis (deg)



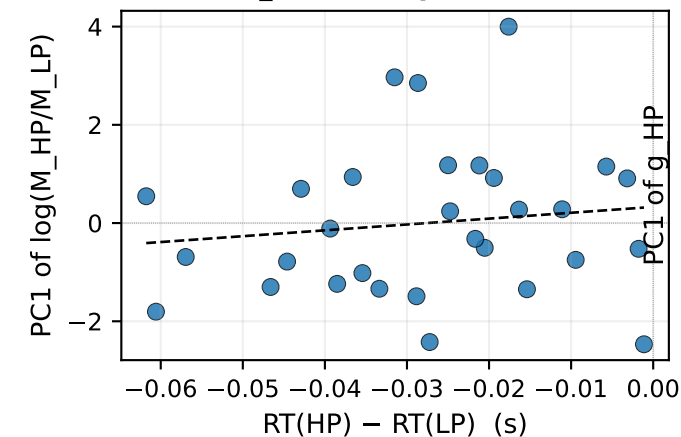
(f) Correlations with behavior

Subject-level co-variation between AF parameters and distractor RT cost.

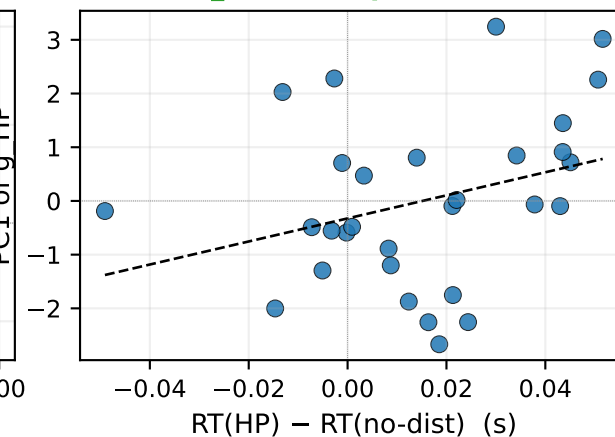
(f) Correlations with behavior (N = 30 subjects)



targeted HP-suppression test
 $r_s = +0.15$, $p = 0.439$



overall HP-AF gain $\times$ HP cost
 $r_s = +0.37$, $p = 0.046^*$



PCA across ROIs collapses 8 weak per-ROI signals into one subject score. The TARGETED test (HP-vs-LP $\times$ HP-vs-LP) is null with $N=30$. The omnibus $g_{HP} \times$ HP-cost test reaches *.

Summary

1. Naive distance-from-HP analyses are biased by Jensen / noise. The pattern, not the magnitude, is what tests the model.
2. Joint AF + PRF model (4 ring AFs, AF+ formulation, single Gaussian voxel kernel) fits per (subject, ROI) at the BOLD level.
3. HP-specific suppression survives in early-to-ventral cortex:

V2	Wilcoxon $p = 0.0002$	80% of 30 subjects $g_{HP} < g_{LP}$
V3	Wilcoxon $p = 0.0002$	80%
V3AB	Wilcoxon $p = 0.05$	
V0	Wilcoxon $p = 0.003$	80%
4. Per-ROI predicted shift vector fields show what each area is doing – see (d) for the spatial pattern at each ROI's empirical median parameters.

Caveats:

- Model still has identifiability issues ($\sigma_{AF} \times |gain|$ trade-off in mid-tier ROIs) but the rank-based HP-vs-LP test survives.
- PRED side uses single Gaussian voxel kernels; OBS side uses DoG (model 4). Apples-to-apples Gaussian conditionwise refit is in flight on cluster.